

SURVEY

From Measurement to Monitoring: Evaluating Privacy Risks in Carbon Footprint Tools

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ABSTRACT The increasing urgency of climate change mitigation has accelerated the adoption of tools to measure and monitor greenhouse gas emissions at individual, organizational, and infrastructural levels. While these tools enhance the accuracy and accountability of emissions measurement, they also raise privacy concerns due to the sensitive data they collect, the granularity of monitoring, and the potential for secondary use. Despite their growing adoption, no prior study has systematically evaluated the privacy implications of carbon footprint tools across categories or examined how data practices differ between consumer calculators, enterprise systems, and emerging Internet of Things (IoT), Artificial Intelligence (AI), and blockchain solutions. This study addresses this gap by conducting a structured, cross-category privacy evaluation of carbon footprint tools, reviewing 46 tools from academic and industry sources. A novel five-dimensional privacy risk framework, encompassing data type, granularity, storage, processing, secondary use, and user agency, is developed and applied across five tool categories: individual calculators, corporate platforms, infrastructure tools, IoT/AI/blockchain systems, and research tools. The findings reveal pronounced privacy-accuracy trade-offs: tools offering higher measurement precision often require more intrusive data collection. The study also identifies category-specific least- and highest-risk tools, showing that simple household calculators present low risk. In contrast, enterprise platforms and IoT/blockchain systems exhibit the highest exposure due to sensitive operational, financial, and real-time data flows. The study is limited by its reliance on literature and public documentation, suggesting the need for empirical field studies and technical audits. Future research should expand on these findings by investigating user perceptions, legal frameworks, and privacy-preserving design strategies for sustainable digital tools.

INDEX TERMS Carbon footprint, data protection, greenhouse gas, privacy risks, risk evaluation, sustainability.

I. INTRODUCTION

Measuring and tracking carbon footprints has become an essential practice in both individual and organizational contexts [1]. With increasing global attention to sustainability, climate action, and the enforcement of reporting regulations, such as the EU Corporate Sustainability Reporting Directive (CSRD), companies and individuals alike require reliable methods for quantifying their greenhouse gas emissions [2]. This demand has led to the development of a diverse range of carbon footprint tools, spanning from simple online

calculators for personal use to advanced, AI-powered platforms integrated with IoT sensors and blockchain-based verification systems [3].

These tools differ widely in terms of scope, accuracy, usability, and privacy implications. Some tools, such as consumer-facing calculators (e.g., World Wildlife Fund (WWF), CoolClimate, MyClimate), are designed to raise awareness and encourage lifestyle changes, while others (e.g., Greenhouse Gas Protocol, One Click Life Cycle Assessment (LCA), Embodied Carbon in Construction Calculator (EC3)) serve professional and industrial applications, supporting compliance, Supply Chain Management (SCM), and infrastructure-level assessments [4], [5]. At the same

The associate editor coordinating the review of this manuscript and approving it for publication was Sangsoo Lim¹.

time, research-oriented tools (e.g., CodeCarbon, Carbon Tracker, FootPrinter) help quantify the environmental cost of computational processes, reflecting the growing impact of digital technologies on climate change [6], [7]. Furthermore, advanced systems are increasingly integrating AI, blockchain, and federated learning to enhance the precision of emissions monitoring and safeguard sensitive business and personal data [8], [9]. This reflects a broader trend toward privacy-preserving, data-driven sustainability tools, as privacy risks have become an essential consideration alongside the accuracy of carbon measurement [10], [11].

Carbon footprint tools are now widely used by individuals, companies, and public institutions to measure and report greenhouse gas emissions [1]. While many studies examine how accurate and effective these tools are, much less attention has been given to the privacy risks created by their data collection practices [4]. There is no clear and structured approach to compare how different types of carbon footprint tools handle sensitive data, such as personal behavior, financial records, or operational information [5]. As newer tools increasingly rely on technologies like IoT, artificial intelligence, and blockchain, the amount and detail of data being collected continue to grow, raising important privacy concerns [2]. This lack of systematic evaluation makes it difficult for users, organizations, and policymakers to understand and manage privacy risks when adopting carbon footprint tools.

While carbon footprint tools have become indispensable for improving the accuracy, transparency, and accountability of emissions measurement, their growing sophistication raises significant concerns about data privacy and security [12], [13]. Many of these tools rely on sensitive forms of data collection, including household energy bills, financial transactions, transportation patterns, GPS tracking, or detailed supply-chain records [14], [15]. The introduction of IoT-based systems, with more than 1.06 billion smart meters already deployed worldwide and projected to exceed 3.4 billion by 2033, exemplifies how advances in data collection enhance precision but simultaneously increase the risks of intrusive monitoring and unintended data exposure [16]. The privacy implications extend beyond individuals to corporations and entire industries [17]. Enterprise-grade platforms, such as those offered by SAP, Microsoft, Salesforce, and Oracle, integrate carbon accounting directly with Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and supply-chain management systems, requiring the collection and processing of highly granular operational and financial data [18], [19]. Although enterprise carbon datasets are primarily collected for internal reporting and compliance, privacy and confidentiality risks arise when such data are shared with third-party ESG platforms, auditors, or regulators, or disclosed through benchmarking and certification schemes. Prior work shows that longitudinal or disaggregated operational data can enable inference of production volumes, supplier relationships, capacity

utilization, and efficiency trends, even when explicit financial metrics are not disclosed [20], [21]. The GDPR explicitly recognizes the risk of “loss of confidentiality of trade secrets or other economically sensitive information” resulting from data processing or disclosure [20]. Recent studies on blockchain-based sustainability reporting further demonstrate how transparent emissions records can unintentionally expose competitive intelligence through pattern analysis [22], [23], [24]. This is particularly salient as reporting obligations expand. According to the Morgan Stanley Capital International (MSCI) report, over 40% of public companies globally now disclose Scope 3 emissions, while in regions like Europe and Japan, disclosure rates approach 90% [25]. Because Scope 3 data capture indirect emissions across entire value chains, they involve contributions from suppliers, distributors, and even customers, creating additional layers of privacy vulnerability [26].

Although carbon footprint tools have been widely examined in terms of their technical accuracy, lifecycle modeling capabilities, and role in sustainability reporting, relatively little research has investigated their privacy implications [27], [28]. Scholarly work tends to emphasize methodological improvements in carbon accounting, such as refining emission factors, integrating Machine Learning (ML), or linking with LCA databases [29]. While many studies emphasize the accuracy and scalability of digital carbon accounting tools, they often overlook how large-scale data integration and secondary inference risks may compromise organizational confidentiality, even when data are stored securely and processed in compliance with industry standards [29]. This gap is particularly evident in enterprise sustainability platforms developed by vendors such as Microsoft, SAP, IBM, Oracle, and Salesforce, not due to inadequate security controls, but because these platforms aggregate high-granularity operational, financial, and supply-chain data across multiple business functions [22], [30]. Such cross-domain aggregation can enable indirect inference of sensitive information, including supplier dependencies, production efficiencies, or strategic priorities, especially when sustainability datasets are reused for reporting, benchmarking, or third-party disclosures beyond their original purpose [30], [31], [32]. While these systems dominate industry practice and are central to Environmental, Social, and Governance (ESG) reporting, they are rarely subject to independent academic scrutiny regarding their privacy implications. Similarly, innovative frameworks based on IoT, AI, and blockchain technologies are often presented as solutions to improve carbon accounting accuracy, but their privacy risks remain underexplored [33], [34]. This study addresses the gap by conducting a comprehensive, targeted literature review of both academic sources and industry documentation. Specifically, the study addresses the following research question:

RQ: How do different categories of carbon footprint tools vary in terms of data collection practices and associated privacy risks?

This study contributes to theoretical understanding by positioning privacy as a core dimension of sustainability measurement, offering practical insights for policymakers, developers, and organizations on how to design and adopt carbon footprint tools responsibly. First, it conducts a structured investigation of carbon footprint tools across five categories: individual or household calculators, corporate or organizational tools, infrastructure or city-focused applications, IoT, AI, blockchain frameworks, and experimental research tools. Second, a five-criteria framework, covering the type of data collected, granularity, storage and processing methods, potential for secondary use, and user agency, was applied to assess privacy risks consistently across categories. Third, it identifies the least and most risky tools in each category. Tools including WWF Footprint Calculator, GES 1point5, EC3, Zero-Knowledge Succinct Non-Interactive Argument of Knowledge (zk-SNARK), and Carbon Tracker have the lowest privacy risk, while tools including Doconomy, Carbon Analytics Dashboards, City Energy Analyst, IoT Sensors/Smart Meters, and CO2web Observatory show the highest privacy risk.

The research highlights privacy exposure patterns for each category, showing which tools are most and least invasive. It reveals trade-offs between measurement accuracy and privacy, with higher accuracy often needing more intrusive data collection. These findings aid stakeholders in making informed decisions and underscore the importance of privacy-aware design in future sustainability tools that incorporate risk factors. The study summarizes as:

- Defined privacy as a core and underexplored dimension of carbon footprint measurement and monitoring systems.
- Conducted a structured, cross-category investigation of carbon footprint tools spanning five distinct groups.
- Developed a five-criterion privacy assessment framework and applied it consistently across all categories.
- Identified category-specific patterns of privacy exposure, highlighting both the least and most privacy-invasive tools within each group.

We begin in Section II by providing background on carbon footprint tools and establishing the significance of their privacy implications. Section III then outlines our research methodology. In Section IV, we present the carbon footprint tools. Section V assesses the privacy risks associated with these tools. Section VI expands on these findings by discussing their broader implications, acknowledging the study's limitations, and suggesting directions for future work. Finally, we conclude Section VII by summarizing the paper's overall contributions.

II. BACKGROUND

A. CLIMATE CHANGE AND THE NEED FOR CARBON ACCOUNTING

Climate change remains one of the defining challenges of the twenty-first century, necessitating urgent collective action to

mitigate greenhouse gas emissions. International agreements such as the Paris Agreement and subsequent Net Zero pledges have set ambitious targets for reducing global warming to below 1.5°C [25]. Achieving these goals requires reliable measurement, monitoring, and reporting of emissions across sectors and scales [35]. Carbon accounting, the process of quantifying and reporting carbon footprints, has therefore become a critical enabler of climate action, providing the data foundation for emissions reduction strategies, policy interventions, and corporate sustainability initiatives [36].

Corporate disclosure practices illustrate this growing importance. A PwC and NUS Business School study of nearly 700 Asia-Pacific companies revealed that 53% have established net-zero targets and 63% disclose Scope 3 emissions, which account for indirect emissions across value chains [37]. Similarly, a Persefoni global survey found that 73% of companies disclose Scope 1 and 2 emissions, but only 26% share Scope 3 data, highlighting a persistent gap in value-chain transparency [38]. In Europe and Japan, disclosure rates are higher, with nearly 90% of MSCI ACWI index companies reporting Scope 3 emissions, compared to a slower uptake in the U.S. market, where companies are only now approaching 90% disclosure for Scopes 1 and 2 [39]. These statistics demonstrate how carbon accounting has become a regulatory and reputational imperative, particularly as investors, regulators, and consumers demand greater accountability. Yet, despite progress in reporting, inconsistencies across regions and reluctance to disclose full Scope 3 data indicate that both the accuracy and the completeness of carbon footprint measurement remain contested [40]. This underscores the importance of developing tools that can both enhance measurement accuracy and protect sensitive data in the process [41].

B. CARBON FOOTPRINT TOOLS: CATEGORIES AND APPLICATIONS

Carbon footprint tools have proliferated in response to the growing demand for carbon accounting across individuals, organizations, and cities [42]. These tools can be broadly classified into five categories, each serving different purposes and user groups. At the individual and household level, online calculators such as those offered by WWF or CoolClimate allow users to input data on lifestyle choices, transportation modes, dietary patterns, and energy consumption to estimate their carbon footprints. These tools primarily serve an educational purpose, encouraging awareness and voluntary behavioral changes [43].

At the corporate and organizational level, enterprise platforms like the Greenhouse Gas Protocol tools, Microsoft Cloud for Sustainability, and Salesforce Net Zero Cloud offer advanced features, including supply chain data integration, automated reporting, and ESG disclosure management [44], [45]. These systems are crucial for companies striving to comply with regulations and meet investor expectations for transparent carbon reporting. Indeed, industry statistics reflect their importance: according to Manifest Climate, 76%

of firms disclose Scope 1 and 2 emissions, while only 55% report Scope 3, reflecting the critical role of enterprise tools in expanding coverage to the most complex emissions categories [46]. On the other hand, infrastructure and city-focused applications, such as One Click LCA or City Energy Analyst, enable planners to assess embodied carbon in materials and design low-carbon urban systems. These applications are crucial in the context of rapid urbanization, where cities are responsible for more than 70% of global CO₂ emissions [46].

IoT, AI, and blockchain-based solutions represent some of the most technologically advanced approaches. These systems leverage real-time sensor data, ML models, and distributed ledgers to provide precise and transparent emissions accounting [47]. Europe alone has already reached ~60% electricity smart meter penetration [48]. These tools significantly enhance data availability but also introduce complex privacy challenges due to their reliance on continuous, granular monitoring. Finally, experimental and research-oriented tools, such as CodeCarbon, ML CO₂ Impact, or Carbon Tracker, provide emissions benchmarks for computing, artificial intelligence workloads, or academic research [49]. While these tools often involve fewer sensitive data, they play a critical role in fostering awareness and methodological innovation within the research community [50]. Together, these categories show the diversity of carbon measurement approaches, ranging from voluntary lifestyle calculators to enterprise-grade, data-intensive platforms. However, this diversity also means that tools vary widely in their data requirements and privacy implications [34].

C. DATA COLLECTION AND PRIVACY CONCERNS IN CARBON TOOLS

Despite their sustainability goals, carbon footprint tools introduce substantial privacy risks due to the types of data they collect. At the individual level, calculators often require sensitive personal information, such as household energy bills, commuting routines, or dietary preferences [51]. More advanced tools, such as Doconomy, link directly to financial transaction data, while others use GPS tracking to monitor travel behavior [52]. These practices risk exposing intimate lifestyle details and raise concerns about data profiling, surveillance, and secondary use [53]. The adoption of IoT-based carbon measurement intensifies these concerns. With over 1.06 billion smart meters deployed globally, households and organizations are generating increasing amounts of real-time energy consumption data [16]. While valuable for precision in carbon accounting, such granular data can also reveal patterns of household occupancy, industrial production cycles, or operational efficiency, create risks of misuse if access by third parties [54].

Corporate and organizational tools also pose distinct challenges. Platforms like SAP, Microsoft, and Salesforce integrate carbon accounting with ERP and CRM systems. This integration involves highly sensitive data, including supplier records, production processes, and financial information. A Manifest Climate study of 117 large firms revealed

that while 76% disclosed Scope 1 and 2 emissions, only 55% disclosed Scope 3 emissions, suggesting that companies remain selective about the operational and supply-chain data they publicly reveal [46]. Such selectivity reflects not only methodological complexity but also privacy and competitive concerns, as disclosure of detailed Scope 3 data can expose strategic relationships or trade secrets. Blockchain-based carbon tools present a different challenge. By design, blockchain emphasizes transparency and immutability, which is beneficial for accountability but problematic for privacy. Records stored on public or semi-public blockchains may inadvertently expose identifiable operational or product lifecycle data, with little recourse for removal or anonymization [55].

III. METHODOLOGY

The methodology adopted in this study is grounded in a comprehensive and targeted literature review, designed to explore and evaluate the privacy risks associated with carbon footprint tools. This approach enabled the authors to synthesize existing scholarly knowledge, incorporate insights from professional practice, and identify critical research gaps in the intersection of carbon measurement and privacy [56]. Unlike a systematic review, which follows rigid inclusion protocols, this study employed an exploratory and selective review strategy, focusing on recent, thematically relevant contributions that align directly with the scope of carbon footprint technologies and their privacy implications [57]. The objective was not only to map the state of the art but also to classify and evaluate the privacy risks of tools across multiple categories, using a consistent set of evaluation criteria. Consequently, the study does not include technical audits, source-code inspections, or penetration testing of the tools themselves.

The literature review primarily drew on peer-reviewed journal articles, conference proceedings, and high-impact academic sources, covering diverse domains such as environmental studies, engineering, and information systems [58]. However, because enterprise-grade carbon management platforms (e.g., Microsoft Cloud for Sustainability, SAP Sustainability Control Tower, IBM Envizi) are underrepresented in scholarly research, the review also included professional and corporate websites to capture their technical descriptions and implementation details. This dual focus allowed the study to bridge the gap between academic analysis and industry practice, ensuring that both experimental research tools and large-scale enterprise solutions were considered.

To ensure breadth and depth of coverage, all well-known academic databases were systematically searched, including Scopus, Web of Science, IEEE Xplore, and ScienceDirect, complemented by supplementary searches in Google Scholar to capture emerging work. The search strategy employed Boolean operators to maximize relevance: “Carbon” OR “CO₂” AND (“Footprint” OR “Measure” OR “Level”) AND (“Privacy” OR “Confidentiality” OR “Integrity”) AND (“risk” OR “concerns” OR “Impact” OR “Effect”). This combination was designed to capture studies examining not only the technical functioning of carbon footprint

TABLE 1. Inclusion criteria and number of selected studies.

Criterion	Description	
Language	Publications in the English language.	
Database	Sourced from all major academic databases and professional websites.	
Publication type	Conference proceedings, book chapter, Journal's articles, websites.	
Typical relevance	Focus on carbon footprint tools.	
Time frame	Publications dated until September 2025.	
Selected articles for the final analysis Total= 72 articles	Before 2019 Number: 8	[54, 60-66]
	Year: 2019 Number: 2	[67, 68]
	Year: 2020 Number: 5	[31, 50, 53, 69, 70]
	Year: 2021 Number: 2	[23, 49]
	Year: 2022 Number: 4	[42, 43, 45, 71]
	Year: 2023 Number: 8	[22, 35, 36, 40, 41, 51, 72, 73]
	Year: 2024 Number: 19	[1, 2, 4-7, 9, 12, 13, 18, 26, 30, 47, 74-79]
	Year: 2025 Number: 24	[3, 10, 11, 14, 15, 17, 27-29, 33, 34, 44, 52, 55, 80-89]

tools but also their socio-ethical and privacy-related implications [59]. Following the retrieval process, inclusion and exclusion criteria were applied to refine the dataset. Table 1 summarizes these criteria, ensuring that only studies with direct relevance to carbon footprint tools and privacy impacts were analyzed. The final dataset, complemented by industry documentation, served as the empirical basis for the privacy evaluation framework developed in this study, which classifies tools into low, medium, or high risk across five dimensions: type of data collected, granularity, storage and processing, potential for secondary use, and user agency.

The initial database search yielded 379 records. After removing duplicates and non-English publications, the dataset was reduced to 39 studies. A two-stage screening process was then employed to ensure alignment with the research objectives. In the first stage, articles were screened based on titles and abstracts, enabling rapid elimination of studies unrelated to carbon footprint tools or privacy. This step narrowed the pool to 119 studies considered suitable for full-text investigation. In the second stage, a full-text review was conducted to refine the dataset. At this stage, studies were excluded if they (a) addressed subject areas outside the scope

of carbon footprint measurement, (b) lacked scholarly credibility by not being peer-reviewed, or (c) focused exclusively on technical aspects of carbon accounting without considering privacy-related dimensions. This refinement process resulted in a final selection of 72 articles deemed relevant for in-depth analysis [59].

To ensure methodological rigor, each selected study was independently reviewed by multiple researchers. To enhance methodological rigor, the privacy risk evaluation followed a structured multi-rater classification process. Three researchers independently assessed each carbon footprint tool across five privacy dimensions: data type, data granularity, storage and processing, potential for secondary use, and user agency. Each dimension was rated using a three-level ordinal scale (low, medium, high), consistent with GDPR principles and ISO/IEC 29100. Inter-rater reliability was assessed using Cohen's kappa, a statistic widely applied for categorical agreement beyond chance [90], [91]. The analysis yielded substantial agreement across classifications ($\kappa = 0.78$), indicating strong consistency among reviewers. Disagreements primarily occurred in hybrid tools with optional data integrations and were resolved through structured deliberation referencing predefined evaluation criteria [91]. Final classifications were assigned only after a unanimous consensus was reached.

To minimize subjectivity and bias, findings were subsequently discussed in collaborative sessions, where differences in interpretation were reconciled through consensus-building. Only those studies that achieved unanimous agreement among reviewers on their relevance and quality were retained for the final synthesis [56]. This iterative and collaborative approach ensured that the final dataset provided a robust foundation for evaluating privacy risks across carbon footprint tools. It also strengthened the reliability of the synthesis by combining independent reviewer judgment with group deliberation, aligning with best practices in qualitative review methodologies. Furthermore, to address the proprietary nature of several enterprise tools, each classification was accompanied by a confidence rating reflecting the level of auditability and disclosure (Table 9).

IV. CARBON FOOTPRINT TRACKING TOOLS

In this paper, we've mapped the tools into five major categories and analyzed their applications, benefits, limitations, and privacy aspects. These categories are: individual and household carbon calculators, corporate and organizational tools, infrastructure, building, and city-level tools, IoT, AI, blockchain, and experimental and research-oriented tools. Figure 1 summarizes the 46 carbon footprint tools identified and evaluated in this paper.

A. INDIVIDUAL AND HOUSEHOLD CARBON CALCULATORS

Tools in this category are typically web or mobile applications that require users to input personal lifestyle data, such

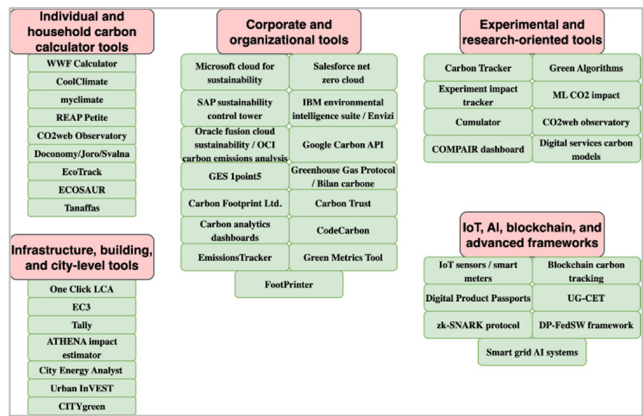


FIGURE 1. Carbon footprint tools identified in this study.

as energy consumption, travel patterns, diet, and purchasing habits. Some newer tools (e.g., EcoTrack, ECOSAUR, Tanaffas) integrate advanced features such as real-time data collection, gamification, or Application Programming Interface (API) links to financial and shopping data. These tools are typically straightforward and are used to raise awareness, educate individuals, encourage behavioral change, and offer options for carbon offsetting. However, the accuracy of these tools is limited due to reliance on self-reported data and generic emission factors. Banking-integrated apps (e.g., Doconomy, Joro, Svalna) are more accurate but raise serious privacy concerns, as they rely on sensitive financial data. Similarly, GPS-based tracking or IoT-linked calculators can intrude on personal privacy by continuously monitoring movements or consumption. Moreover, these tools generally require extensive personal lifestyle data, which creates significant privacy risks. Table 2 summarizes each of these tools.

1) WWF FOOTPRINT CALCULATOR

The WWF Footprint Calculator is a widely recognized online tool that estimates an individual’s ecological footprint across key lifestyle domains, including housing, transportation, diet, and consumption patterns, through a straightforward, questionnaire-based interface [85]. Developed with public engagement and education as primary goals, it serves as a powerful instrument in sustainability awareness campaigns, helping users visualize the environmental impact of their daily choices in an accessible and intuitive manner [68]. Its strengths lie in its user-friendly design, clear visual feedback, and minimal cognitive burden, making it particularly effective for non-expert audiences seeking to understand their personal contribution to global resource use [85].

2) COOLCLIMATE CALCULATOR

The CoolClimate Calculator, developed by researchers at UC Berkeley, is a sophisticated online tool designed to estimate household-level carbon footprints with greater granularity than most public-facing calculators [67]. It prompts users to provide detailed inputs on housing energy use (including

TABLE 2. Individual and household carbon calculators’ tools.

Tool	Applications	Strengths
WWF Calculator [68, 85]	Awareness, education	Simple, user-friendly
CoolClimate [67, 85]	Education, household analysis	Academic credibility, detailed
MyClimate [72]	Awareness, voluntary offsets	Linked to projects
REAP Petite [61]	Household-level analysis	Higher accuracy
CO2web Observatory [50]	Education, comparisons	Combines product and personal
Doconomy/Joro/Svalna [52, 53]	Consumer tracking	High accuracy via transactions
EcoTrack [76]	Personal coaching	Recommendations, analytics
ECOSAUR [81]	Real-time tracking	Dynamic, continuous
Tanaffas [78]	Gamified awareness	Engaging, interactive

heating, cooling, and electricity sources), transportation habits (such as vehicle type, mileage, public transit usage, and air travel frequency), dietary patterns, and consumption behaviors, including spending on goods, services, and entertainment, allowing for a more nuanced and scientifically grounded assessment of personal emissions [85]. Backed by rigorous academic research and continuously refined using life-cycle analysis models, the calculator not only delivers personalized footprint estimates but also integrates policy scenarios that let users explore how systemic changes, like switching to renewable energy or adopting electric vehicles, could reduce their impact, making it especially valuable for educational outreach and behavior-change initiatives aimed at informed decision-making [67].

3) MYCLIMATE

MyClimate is a non-profit online carbon footprint calculator developed with the dual mission of raising personal environmental awareness and facilitating voluntary participation in certified carbon offset programs [72]. Designed for individual users seeking to understand and mitigate their climate impact, the tool estimates emissions across key lifestyle categories, including travel, home energy use, and consumption, using a streamlined and intuitive interface that prioritizes accessibility over complexity. Its primary strength lies in its seamless integration with verified carbon offset projects, enabling users to immediately invest in reforestation, renewable energy, or community-based initiatives after receiving their footprint estimate, thereby transforming awareness into tangible action [72].

4) REAP PETITE

REAP Petite is a sophisticated, consumption-based carbon footprint calculator developed to deliver highly accurate, household-level emissions estimates by analyzing detailed spending and lifestyle patterns rather than relying on broad

averages [61]. Unlike simpler tools that ask general questions about travel or diet, REAP Petite requires users to input granular data, such as itemized purchases, brand-specific consumption, utility usage, commuting habits, and even food types and quantities, leveraging life-cycle inventory databases to map every dollar spent to its associated greenhouse gas emissions [61]. This level of detail enables unprecedented precision, allowing researchers and policy-makers to identify hidden hotspots in household footprints and compare environmental impacts across socioeconomic groups with scientific rigor [61]. Its strength lies in its ability to uncover subtle behavioral drivers of emissions, making it invaluable for academic studies and targeted sustainability interventions.

5) CO2WEB OBSERVATORY

The CO2web Observatory is an online platform that combines individual carbon footprints with a detailed product-level emissions database, allowing users to assess both their personal impact and the environmental costs of everyday goods, from food and clothing to electronics and transportation [50]. Designed for education and awareness, it enables comparisons between products based on life-cycle data, helping users make informed, low-carbon choices by revealing hidden emissions associated with consumption. Its unique strength lies in connecting macro-scale supply chain insights with micro-level behavior, turning abstract carbon numbers into tangible, relatable decisions [50]. While users manually enter personal data, avoiding automated tracking or external integrations, the linkage between individual inputs and product databases raises medium privacy concerns. Specifying brands, purchase frequencies, or spending categories can inadvertently expose sensitive lifestyle patterns, income levels, or ethical preferences (e.g., organic diets, sustainable fashion).

6) DOCONOMY, JORO, AND SVALNA

Doconomy, Joro, and Svalna are mobile apps that calculate personal carbon footprints by automatically analyzing financial transaction data, linking credit card purchases, bank transfers, and subscription payments to emissions databases to estimate the climate impact of every expense [53]. Their key strength lies in seamless, real-time tracking: users gain highly accurate, granular insights into how spending on flights, dining, retail, or energy bills contributes to their footprint, without manual input or estimation bias [52]. This automation drives effective behavior nudging, encouraging lower-carbon choices through immediate feedback. However, this convenience comes at a very high privacy cost [53].

7) ECOTRACK

EcoTrack is a mobile app that estimates personal carbon footprints by collecting detailed lifestyle inputs, such as commuting habits, diet, energy use, and consumer spending, and delivers personalized, AI-driven recommendations to support sustainable behavior change [76]. Designed for

personal tracking, education, and lifestyle coaching, it leverages modern cloud infrastructure, specifically Google Cloud, to analyze user data and generate real-time, tailored advice, enhancing engagement through dynamic feedback loops. Its strength lies in its seamless integration of ML and scalable analytics, enabling accurate, adaptive guidance without requiring manual recalculations [76]. However, this convenience comes at a high privacy cost: all sensitive personal data is stored and processed on external third-party servers, exposing users to risks of data breaches, unauthorized access, or opaque data-sharing practices.

8) ECOSAUR

ECOSAUR is a cutting-edge mobile app that revolutionizes carbon footprint tracking by leveraging real-time data streaming technologies, such as Apache Kafka and Spark, to continuously collect, process, and analyze user lifestyle inputs, including location, transportation modes, energy use, and purchasing behavior [81]. Unlike static calculators that rely on periodic surveys, ECOSAUR delivers dynamic, near-instantaneous feedback, enabling users to see the immediate climate impact of their actions, from choosing public transit over driving to opting for plant-based meals [81]. This real-time responsiveness makes it uniquely effective for fostering sustained engagement and behavioral nudges through contextual awareness [81].

9) TANAFFAS APP

Tanaffas is a highly immersive, gamified carbon footprint app that transforms sustainability into an interactive experience by integrating GPS tracking, real-time shopping receipt scanning, augmented reality object detection for product emissions, and social challenges to drive user engagement [78]. Designed primarily for younger demographics, it turns daily choices, such as opting for a bus over a car or purchasing local produce, into points, badges, and leaderboards, fostering competition and community-driven behavior change. Its strengths lie in its captivating design: AR scans identify products and instantly display their carbon cost, while GPS logs travel patterns without manual input, creating a seamless, playful interface that educates through immersion [78]. Yet this innovation comes at an extreme privacy cost.

B. CORPORATE AND ORGANIZATIONAL TOOLS

These category tools are designed for organizational sustainability reporting, energy and emissions monitoring, or specialized assessments in computational and IT contexts. These tools can be utilized for corporate carbon accounting, compliance with sustainability standards, organizational reporting, internal benchmarking, and green IT initiatives. The strengths of these tools are demonstrated through numerous standards, which are based on frameworks such as the Greenhouse Gas Protocol and ISO 14064, thereby ensuring global credibility. Tools like GES 1point5 emphasize privacy by design, particularly for sensitive research contexts. Computational tools (CodeCarbon, EmissionsTracker, Green Metrics Tool,

FootPrinter) highlight the rising awareness of digital carbon footprints, especially in ML and cloud computing. However, many organizational tools are resource-intensive, requiring detailed operational data and technical expertise. Spreadsheets (such as the Greenhouse Gas Protocol and Bilan Carbone) lack automated privacy safeguards. API-based tools (e.g., Google Carbon API) are limited to corporate users and may require access to sensitive organizational data. Also, while organizational tools often focus on aggregated data, they may still expose confidential business information, especially when reporting at granular levels. Privacy-preserving protocols (e.g., selective disclosure, zero-knowledge proofs) are emerging to mitigate this risk. Table 3 summarizes each of these tools.

1) MICROSOFT CLOUD FOR SUSTAINABILITY

Microsoft Cloud for Sustainability provides powerful enterprise ESG analytics with seamless integration into Azure and Dynamics, automating Scope 1–3 reporting and product lifecycle tracking [92]. Its strength lies in scalability and regulatory compliance. Still, its privacy impact is high: it centralizes sensitive operational data, energy use, supply chains, and manufacturing logs on Microsoft's cloud, creating significant exposure to breaches or re-identification [92]. While secure, the platform lacks transparency in data usage and offers no local processing, forcing enterprises to trust a single vendor with proprietary insights. Convenience comes at the cost of autonomy: even anonymized data can reveal competitive strategies. It enables compliance, but at the risk of corporate surveillance by proxy [92].

2) SALESFORCE NET ZERO CLOUD

Salesforce Net Zero Cloud is an ESG platform embedded within the Salesforce ecosystem, enabling enterprises to track emissions, engage suppliers, and automate disclosures via integrated CRM and supply chain data [93]. Its strengths lie in seamless customer and vendor relationship mapping, automated reporting workflows, and stakeholder dashboards that align with frameworks such as the Greenhouse Gas Protocol [93]. While secure, this creates risks of competitive exposure, re-identification through network analysis, or unintended data sharing across Salesforce's broader app ecosystem. Enterprises gain efficiency but surrender control over confidential supply chain insights [93].

3) SAP SUSTAINABILITY CONTROL TOWER

SAP Sustainability Control Tower unifies real-time emissions data from SAP-ERP, SCM, and external systems, shifting reporting from estimates to actuals across Scope 1–3, energy, water, and waste [94]. Its strength lies in deep native integrations, granular role-based access controls, and audit-ready compliance workflows tailored for regulated industries [94]. Even if enterprise platforms have governance features such as access controls and role-based permissions to prevent competitive profiling, this profiling can still occur through inference without direct access to the data [2]. The

TABLE 3. Corporate and organizational tools.

Tool	Applications	Strengths
Microsoft cloud for sustainability [92]	Enterprise ESG tracking; Scope 1–3 reporting; product lifecycle analysis; governance & compliance.	Strong MS ecosystem integration (Azure, Dynamics), automated connectors, regulatory & audit support.
Salesforce net zero cloud [93]	Corporate footprint; supplier engagement; ESG reporting; stakeholder dashboards.	Deep CRM/supplier integration, automation, prebuilt reporting flows.
SAP sustainability control tower [94]	Value-chain emissions, audit-ready reporting, operational sustainability decisions.	Native SAP integrations (ERP/SCM), configurable metrics, governance & role controls.
IBM environmental intelligence suite / Envizi [95]	Enterprise emissions management; climate risk planning; supply-chain emissions.	Strong risk, geospatial analytics, supply-chain integration, enterprise reporting.
Oracle fusion cloud sustainability / OCI carbon emissions analysis [96]	Corporate sustainability reporting; cloud workload carbon analysis; procurement/supply chain metrics.	Integration with Fusion ERP/EPM, OCI carbon analysis, Sustainability ledger / audit trails.
Google Carbon API [44]	IT sustainability	Seamless cloud integration
GES 1point5 [45]	Universities, institutions	Privacy safeguards, transparent
Greenhouse Gas Protocol / Bilan Carbone [45, 60]	Corporate reporting, policy	Global standard, comprehensive
Carbon Footprint Ltd. [67]	Corporate reporting, offsetting	Widely used, accessible
Carbon Trust [67]	Corporate management	Trusted partner, established
Carbon analytics dashboards [41]	ESG reporting, oversight	System integration, automation
CodeCarbon [88]	Research, ML, IT	Lightweight, open-source
EmissionsTracker [88]	IT sustainability	Precise regional analysis
Green Metrics Tool [44]	Software diagnostics	Network + power analysis
FootPrinter [79]	Data center policy	Scenario testing

detailed, granular emissions, operational, and supply-chain data aggregation may reveal patterns of production efficiency or supplier structure, although individual records may be protected [3]. Data sovereignty is influenced as sustainability data are generally kept and processed in vendor-managed cloud ecosystems, which often use proprietary models, thus limiting the organization's freedom to decide where and how these data are handled. Therefore, enterprises get the benefits of accuracy and compliance from the vendors, but their autonomy over strategically sensitive sustainability information is diminished [94].

4) IBM ENVIRONMENTAL INTELLIGENCE SUITE/ENVIZI

IBM Environmental Intelligence Suite (with Envizi) provides enterprise-grade emissions accounting, featuring advanced geospatial, weather, and supply-chain analytics, which enables climate risk modeling and Scope 3 tracking across global operations [95]. Its strength lies in integrating diverse data sources, satellite imagery, IoT sensors, and supplier databases to deliver predictive, location-aware insights for strategic decision-making [95]. While robust security measures exist, the platform's depth of integration means sensitive business intelligence is stored and processed by IBM, raising concerns over vendor lock-in and opaque data usage policies. Enterprises gain powerful foresight, but at the cost of ceding control over their most confidential environmental and operational data to a third-party cloud provider [95].

5) ORACLE FUSION CLOUD SUSTAINABILITY

Oracle Fusion Cloud Sustainability, powered by Oracle Cloud Infrastructure (OCI) Carbon Emissions Analysis, enables enterprise-wide Scope 1–3 reporting by integrating emissions calculations directly with ERP, EPM, and cloud infrastructure data, using power consumption metrics from OCI workloads to estimate digital carbon footprints [96] automatically. Its strength lies in seamless, real-time integration with Oracle's ecosystem, automated audit trails, and a dedicated sustainability ledger for compliance [96]. While secure, this centralization creates risk: breaches or insider access could expose competitive supply chain patterns, manufacturing processes, or cloud usage strategies. Even anonymized outputs may be re-identified through behavioral analysis. Enterprises gain precision and automation, but trade data sovereignty for vendor dependency [96].

6) GOOGLE CARBON API

Google Carbon API is a specialized service integrated within Google Cloud that automatically estimates the carbon emissions generated by computational workloads, such as virtual machines, data storage, and serverless functions, by correlating energy usage with real-time grid intensity data across Google's global data centers [44]. Designed for IT and sustainability teams managing cloud infrastructure, it enables organizations to track, visualize, and optimize the environmental impact of their digital operations without manual input or third-party tools [44]. Its key strength lies in seamless, automated integration: by leveraging Google's proprietary infrastructure metrics and renewable energy attribution models, it delivers granular, accurate, and timely emissions insights directly into existing cloud dashboards, empowering companies to make energy-efficient architecture decisions and meet ESG reporting obligations [44]. While the Google Carbon API represents a significant advancement in scalable, automated emissions accounting, its value is inseparable from its dependency on corporate trust in a single provider [44].

7) GES 1POINT5

GES 1point5 is a research-oriented carbon footprint platform designed specifically for academic institutions, research centers, and universities seeking to measure and report organizational emissions with rigorous scientific integrity and uncompromising privacy standards [45]. Unlike consumer-facing apps that track individuals, GES 1point5 focuses on institutional-level data, such as energy consumption, procurement patterns, travel logs, and waste generation, aggregating inputs at the departmental or operational level rather than the personal one [45]. Its defining strength lies in its commitment to privacy-by-design: all data is anonymized from the outset, stripped of Personally Identifiable Information (PII), stored minimally and temporarily, and processed only for aggregated reporting purposes [45]. The tool is fully compliant with GDPR and other stringent data protection frameworks, ensuring no individual's behavior, identity, or private activity can be inferred or reconstructed from the outputs. Methodology is fully transparent, grounded in peer-reviewed life-cycle assessment protocols, and openly documented to support reproducibility and academic trust [45].

8) GREENHOUSE GAS PROTOCOL / BILAN CARBONE

The Greenhouse Gas Protocol and France's Bilan Carbone are foundational methodologies for comprehensive greenhouse gas accounting, widely adopted by corporations, governments, and institutions to measure emissions across Scope 1 (direct emissions), Scope 2 (indirect emissions from energy), and Scope 3 (value chain emissions) sources [45]. Unlike consumer-facing apps, these frameworks operate at the organizational level, requiring detailed internal data on fuel use, electricity consumption, supply chain logistics, business travel, employee commuting, and product lifecycles [60]. Their primary strength lies in global recognition and standardization: they align with ISO 14064, CDP, and SEC climate disclosure requirements, ensuring consistency, comparability, and regulatory compliance across sectors and borders. This makes them indispensable for corporate sustainability reporting, national inventories, and evidence-based policymaking [45]. However, while they do not collect personal identifiers or track individuals directly, their implementation often involves gathering sensitive operational data, such as supplier contracts, production volumes, transportation routes, and employee commuting patterns, that, if mishandled, could expose proprietary business intelligence or reveal workforce demographics and behaviors [60].

9) CARBON FOOTPRINT LTD. CALCULATOR

Carbon Footprint Ltd.'s Calculator is a widely used online platform that serves both individuals and organizations, offering a streamlined interface for estimating emissions alongside integrated carbon offsetting options and consultancy services [67]. Designed to bridge public awareness with corporate responsibility, it enables businesses to report emissions across key scopes while also empowering employees

to calculate personal footprints through simplified questionnaires, making it a popular tool for sustainability engagement programs [67]. Its strengths lie in accessibility and versatility: users can generate quick reports, compare benchmarks, and purchase certified offsets directly through the platform, creating a seamless pathway from measurement to action [67]. However, its privacy impact is medium due to its hybrid structure. While individual users provide only basic lifestyle data, organizational clients must input sensitive operational information, including energy bills, fleet details, supply chain volumes, and employee travel logs, which are uploaded to a third-party cloud infrastructure managed by Carbon Footprint Ltd [67]. This centralization introduces risks of data exposure, unauthorized access, or unintended aggregation of proprietary business metrics.

10) CARBON TRUST TOOLS

Carbon Trust Tools is a comprehensive suite of digital and advisory services developed by the respected UK-based Carbon Trust to support organizations in measuring, managing, and reporting their carbon footprints [67]. Designed primarily for businesses, public institutions, and supply chains, the tools facilitate compliance with international standards, such as ISO 14064 and the Greenhouse Gas Protocol, enabling the accurate calculation of Scope 1, 2, and 3 emissions through tailored templates, sector-specific calculators, and expert-led verification processes [67]. Their primary strength lies in credibility and institutional trust, as they are used by global corporations and government agencies alike. They provide authoritative, audit-ready outputs that enhance sustainability reporting, investor confidence, and policy alignment [67]. Carbon Trust Tools exemplify the norm in professional carbon accounting: precision demands transparency, but transparency demands trust. For enterprises navigating both climate accountability and data governance, using these tools requires clear contractual safeguards, strict access controls, and an understanding that even trusted partners can become vectors for corporate data exposure [67].

11) CARBON ANALYTICS DASHBOARDS

Carbon Analytics Dashboards are enterprise-grade platforms designed to automate corporate carbon footprint tracking by integrating directly with internal financial, ERP, supply chain, and energy management systems. They pull real-time data from accounting software, procurement records, utility meters, and logistics databases to generate dynamic, audit-ready emissions reports [41]. Tailored for ESG disclosure, executive oversight, and regulatory compliance, these dashboards enable organizations to monitor Scope 1–3 emissions at scale, identify hotspots, forecast trends, and align sustainability metrics with financial performance [41]. Their primary strength lies in automation and scalability: by eliminating manual data entry and syncing with existing corporate infrastructure, they deliver unprecedented accuracy, efficiency, and strategic insight for large, complex operations [41]. These dashboards represent the cutting

edge of corporate climate governance, but also its greatest vulnerability [41].

12) CODECARBON

CodeCarbon is an open-source Python library designed to estimate the energy consumption and associated carbon emissions of computational workloads, particularly in AI, ML, and data-intensive research, by monitoring hardware utilization (CPU, GPU, memory) and correlating it with regional grid intensity data [88]. Unlike commercial or cloud-based tools, CodeCarbon runs locally on a user's device or server, requiring no external data transmission, user registration, or cloud dependency [88]. Its primary strength lies in its transparency, accessibility, and alignment with academic and ethical computing values: researchers, developers, and institutions can integrate it directly into training pipelines to track the environmental cost of algorithms in real time, fostering accountability in a rapidly growing sector often criticized for its hidden carbon footprint [88]. CodeCarbon exemplifies how sustainability tools can be both scientifically rigorous and ethically sound: by placing control firmly in the hands of the user and avoiding any form of data extraction, it transforms environmental awareness into a private, self-directed practice, not a monitored activity [88]. In an ecosystem increasingly dominated by corporate tracking platforms, CodeCarbon stands as a vital counter-model: a tool that measures impact without compromising integrity.

13) EmissionsTracker

EmissionsTracker is an open-source Python package designed to estimate the carbon emissions of computational workloads such as data processing, AI training, and high-performance computing, by dynamically integrating real-time regional grid intensity data to calculate electricity-based emissions with high spatial precision [88]. Unlike cloud-based or corporate monitoring tools, EmissionsTracker operates locally on a user's machine or server, requiring no account creation, external logins, or data transmission to third parties [88]. It derives its estimates solely from system-level metrics, including CPU/GPU utilization, runtime, and power consumption, combined with publicly available, geolocated electricity carbon factors (e.g., from the International Energy Agency or Electricity Map), enabling accurate, context-aware assessments that reflect whether a workload runs in a coal-heavy region or a renewables-dominant grid [88].

14) GREEN METRICS TOOL

Green Metrics Tool is an open-source software sustainability platform designed to measure the environmental impact of digital systems by quantifying power consumption, CPU/GPU utilization, memory usage, and network traffic during application execution [44]. Built for developers, researchers, and IT sustainability teams, Green Metrics Tool provides granular, real-time diagnostics on how code efficiency translates into energy use, enabling optimization of

algorithms, containers, and cloud services without requiring changes to infrastructure or reliance on proprietary vendors [44]. Its strength lies in its focused, technical scope: unlike consumer-facing carbon calculators, Green Metrics Tool operates at the system level, analyzing only hardware and network performance metrics through lightweight instrumentation, often running locally on test environments or CI/CD pipelines [44]. The Green Metrics Tool exemplifies how environmental accountability in tech can be achieved without surveillance. By measuring the footprint of code, not the character of users, it turns sustainability into a matter of engineering rigor rather than behavioral tracking [44].

15) FootPrinter

FootPrinter is a simulation-based research tool built on the OpenDC framework, designed to model and analyze the energy consumption and carbon footprint of data centers under varying operational conditions [79]. Rather than collecting real-time user or organizational data, it uses synthetic inputs, such as server configurations, cooling systems, workload patterns, and regional grid carbon intensities, to simulate how different design choices, scaling strategies, or policy interventions impact emissions over time [79]. This makes it an invaluable asset for academic researchers, data center operators, and policymakers seeking to evaluate sustainability scenarios without requiring access to sensitive infrastructure details. Its key strength lies in its ability to test “what-if” hypotheses: for example, comparing the emissions reduction potential of liquid cooling versus air cooling, or assessing the impact of relocating a facility to a region with higher renewable penetration, all without requiring live data from actual facilities [79].

C. INFRASTRUCTURE, BUILDING, AND CITY-LEVEL TOOLS

These category tools are macro-level assessment platforms applied in construction, urban planning, and infrastructure management. Applications: Estimating embodied carbon in construction materials, assessing building designs, urban energy planning, and city-wide sustainability projects. These tools are highly detailed and compliant with standards (e.g., EN 15978, LEED, BREEAM). Tools like EC3 are open access, increasing transparency in materials selection. BIM-integrated tools such as Tally support real-time assessment during design. However, there are also high data demands requiring integration with construction specifications, supply chain data, and BIM systems. Limited applicability to personal or organizational tracking. Additionally, individuals are generally at low risk, as they rely on aggregated technical data rather than personal lifestyle data. However, they may involve sensitive supply chain data (e.g., material sourcing) at the corporate level. Table 4 summarizes each of these tools.

1) One Click LCA

One Click LCA is a specialized LCA platform designed for the architecture, engineering, and construction industries, enabling professionals to quantify the environmental impact

TABLE 4. Infrastructure, building, and city-level tools.

Tool	Applications	Strengths
One Click LCA [71]	Architecture, infrastructure	Automates BIM data, standard-aligned
EC3 [71]	Procurement, design	Transparency, materials DB
Tally [71]	Architecture, certification	Real-time design integration
ATHENA impact estimator [71]	Construction, certification	Comprehensive database
City Energy Analyst [71]	Smart cities, urban planning	Open-source, detailed scenarios
Urban InVEST [71]	Land use, policy	Ecosystem + emissions modeling
CITYgreen [71]	Municipalities, green planning	Nature-based carbon accounting

of buildings and infrastructure projects, from material extraction to end-of-life disposal [71]. Integrated directly with building information modeling software such as Revit and Archicad, it automatically extracts geometric and material data from digital building models, eliminating manual entry and significantly improving accuracy and efficiency [71]. The platform supports global LCA databases and aligns with international standards, such as ISO 14040/44 and environmental product declaration frameworks, making it indispensable for achieving green certifications, including LEED, BREEAM, and DGNB [71]. Its strength lies in its ability to handle complex, project-scale datasets, tracking thousands of materials across structural, mechanical, and finishes systems, with intuitive visualization tools that help designers optimize for low-carbon alternatives [71].

2) EMBODIED CARBON IN CONSTRUCTION CALCULATOR

EC3, developed by the nonprofit Building Transparency, is an open-access digital tool designed to help architects, engineers, and contractors evaluate and reduce the embodied carbon of building materials, which refers to the emissions generated during extraction, manufacturing, transportation, and installation [71]. Unlike tools that track user behavior or personal habits, EC3 operates as a transparent marketplace for environmental data: users input material specifications (e.g., concrete mix, steel grade, insulation type) and select from its vast, crowdsourced database of environmental product declarations, which are independently verified product-level lifecycle assessments [71]. The platform enables real-time comparison of materials, supports procurement decisions aligned with low-carbon goals, and facilitates benchmarking across projects, all while integrating seamlessly into design workflows [71]. Its greatest strength lies in democratizing access to critical sustainability data. By making thousands of environmental product declarations freely available and searchable, EC3 has become the de facto

standard for green construction, empowering teams to meet codes like ASHRAE 189.1 or comply with emerging regulations such as California's SB 253 [71].

3) TALLY

Tally is a BIM-integrated plugin developed for Autodesk Revit, enabling architects and designers to perform real-time LCA of building designs directly within their modeling environment [71]. By automatically extracting material quantities and geometry from the BIM model, Tally calculates embodied carbon and environmental impacts across multiple categories, including global warming potential, acidification, and eutrophication, allowing design teams to compare alternatives and optimize for sustainability during the earliest and most impactful stages of a project [71]. Its key strength lies in seamless integration: unlike post-design reporting tools, Tally embeds environmental feedback into the creative process, empowering architects to make informed, low-carbon decisions without disrupting workflow or requiring manual data entry [71]. It supports compliance with green certification systems, such as LEED, EN 15804, and the 2030 Challenge, making it indispensable for firms committed to climate-responsive design [71].

4) ATHENA Impact Estimator

ATHENA Impact Estimator is a well-established software tool designed to assess the full lifecycle environmental impacts of buildings, including carbon emissions, energy use, water consumption, and waste generation, by analyzing building assemblies, materials, and construction methods [71]. Primarily used by architects, engineers, and sustainability consultants, it enables comprehensive, code-compliant analysis for green certifications such as LEED, BREEAM, and Green Globes, supporting informed decisions in material selection, design optimization, and lifecycle planning. Its strength lies in its robust, peer-reviewed database of building components and regional environmental profiles, allowing users to generate detailed reports based on standardized LCA methodologies (ISO 14040/44) without requiring input of raw manufacturing data [71].

5) CITY ENERGY ANALYST

City Energy Analyst is an open-source software platform designed for comprehensive analysis of urban energy systems, enabling planners and policymakers to model energy consumption, demand profiles, and carbon emissions across entire cities or districts [71]. By integrating building geometry, occupancy patterns, climate data, infrastructure networks, and energy supply sources, City Energy Analyst simulates the performance of neighborhoods under various scenarios, such as retrofitting buildings, deploying district heating, or increasing renewable penetration, making it indispensable for smart city planning, climate action planning, and policy design [71]. Its key strength lies in its high-resolution, systems-level modeling: it moves beyond aggregate city totals to estimate energy use at the building block or

even individual building level, supporting granular interventions that maximize efficiency and equity [71]. The tool embodies a critical tension in urban sustainability: achieving precision requires granularity, but granularity demands responsibility [71].

6) URBAN INVEST

Urban InVEST is an ecosystem service modeling platform adapted for urban environments, designed to help planners and policymakers evaluate the environmental trade-offs between land use decisions and climate outcomes, such as how green space expansion reduces urban heat islands, improves air quality, or sequesters carbon, while also accounting for emissions from infrastructure development [71]. Developed by the Natural Capital Project, it integrates spatial data on vegetation, soil, impervious surfaces, and building density with biophysical models to quantify ecosystem services alongside greenhouse gas impacts at the neighborhood or city scale [71]. Its primary strength lies in its holistic, systems-based approach: rather than isolating carbon emissions, it situates them within broader ecological contexts, revealing co-benefits like flood mitigation, biodiversity support, and human health improvements that inform more resilient urban design [71].

7) CITYgreen

EC3, CITYgreen is a geographic information system-based software tool designed to quantify the carbon storage and sequestration potential of urban forests and green infrastructure, such as trees, parks, street vegetation, and green roofs, by analyzing land cover, tree species, canopy density, and soil characteristics using spatial data layers [71]. Developed for municipalities, urban planners, and environmental agencies, it enables precise mapping of where trees capture and store atmospheric CO₂, helping cities assess the climate benefits of their green assets and prioritize planting or preservation efforts in high-impact areas [71]. Its key strength lies in its visual, map-driven interface that integrates seamlessly with geographic information systems, allowing users to overlay ecological data with demographic, socioeconomic, or infrastructure layers to identify equity gaps in tree coverage and target interventions strategically [71].

D. IoT, AI, BLOCKCHAIN, AND ADVANCED FRAMEWORKS

This is a rapidly growing category. These tools are usually used for real-time monitoring of energy and emissions, smart grid optimization, international carbon trading, secure supply chain monitoring, and privacy-preserving carbon accounting. These tools are at the frontier of innovation. Blockchain provides transparency and immutability, while federated learning and zero-knowledge proofs enhance privacy [8]. Digital twins and IoT sensors enable dynamic, real-time footprint estimation. Digital Product Passports show strong potential in supply chain traceability, particularly for green hydrogen and industrial products. However, they have complexity, cost, and interoperability issues. IoT sensors and

TABLE 5. IoT, AI, blockchain, and advanced frameworks.

Tool	Applications	Strengths
IoT sensors / smart meters [70, 75, 86]	Households, industries, smart grids	High accuracy, dynamic data
Blockchain carbon tracking [22, 80, 84]	Supply chain, carbon credits	Immutable, transparent
Digital Product Passports [77]	Industrial traceability	EU-aligned, transparency
UG-CET [26]	Carbon markets	Decentralized, global
zk-SNARK protocol [55]	Cloud, enterprise	Preserves confidentiality
DP-FedSW framework [82]	Industrial monitoring	AI + federated learning
Smart grid AI systems [75]	Utilities, renewables	Predictive optimization

smart meters generate vast amounts of highly granular data, which raises significant cybersecurity and privacy concerns. Blockchain tools may unintentionally expose sensitive data unless combined with advanced privacy-preserving techniques. Also, they are double-edged, IoT devices and smart meters are highly privacy-intrusive, potentially revealing intimate details of household or corporate activity. Table 5 summarizes each of these tools.

1) IoT SENSORS AND SMART METERS

IoT sensors and smart meters represent a foundational technology for real-time monitoring of energy consumption, indoor air quality (including CO₂ levels), temperature, humidity, and occupancy patterns in homes, commercial buildings, and industrial facilities, enabling dynamic optimization of heating, cooling, lighting, and grid demand response [75]. Integrated into smart grids and home automation systems, they provide unprecedented accuracy in tracking energy use down to the minute, empowering utilities, building managers, and consumers to reduce waste, lower emissions, and enhance efficiency [70]. Their strength lies in their ability to enable predictive maintenance, load balancing, and personalized feedback loops that drive behavioral change at scale. However, this precision comes at an extremely high privacy cost [86]. The granular, continuous data streams generated by these devices, such as when lights turn on or off, when appliances cycle (e.g., refrigerator, washing machine, oven), or how long someone is home, can be reverse-engineered to infer highly intimate details about daily life: sleep schedules, work hours, family composition, medical conditions (e.g., oxygen concentrators or dialysis machines), religious practices (e.g., times of prayer or fasting), vacation patterns, and even political affiliations based on appliance usage tied to media consumption. Referring to political affiliation means that it is an indirect and probabilistic inference rather than an explicitly identified one [86]. Data from smart appliance meters can show the usage patterns

of media devices such as TVs or streaming devices, along with the time [75]. If these are correlated with known content schedules or viewing habits, it may become possible to a small extent behavioral or preference profiling [70]. These kinds of risks are, therefore, mostly attributable to secondary analysis or data reuse, rather than the primary function of carbon monitoring tools. In industrial settings, similar data can expose proprietary production cycles, shift patterns, or operational vulnerabilities [70].

2) BLOCKCHAIN CARBON TRACKING SYSTEMS

Blockchain Carbon Tracking Systems utilize distributed ledger technologies, such as Hyperledger Fabric, Ethereum, or private permissioned chains, to create immutable and transparent records of greenhouse gas emissions across complex supply chains, spanning from raw material extraction to final product delivery [84]. Designed for applications in carbon credit markets, corporate ESG reporting, and regulatory compliance, these systems enable decentralized verification of emissions data, reducing fraud, enhancing auditability, and fostering trust among stakeholders who may not fully trust traditional third-party certifications [80]. Their key strength lies in cryptographic immutability: once an emission event is recorded (e.g., a shipment's fuel use, a factory's energy intake, or a ton of CO₂ offset), it cannot be altered or deleted, ensuring integrity even in adversarial environments [84].

3) DIGITAL PRODUCT PASSPORT

The Digital Product Passport is a blockchain-enabled system designed to track and verify the carbon footprint of industrial products, specifically developed for the hydrogen supply chain, to ensure transparency, traceability, and compliance with emerging regulatory frameworks, such as the EU's Green Deal and Carbon Border Adjustment Mechanism [77]. Each Digital Product Passport acts as a tamper-proof digital identity for a unit of hydrogen, embedding data on production methods (e.g., green, blue, or gray), energy sources, geographic origin, emissions intensity, certification status, and transport logistics, all recorded on a permissioned blockchain accessible to authorized stakeholders across the value chain [77]. Its primary strength lies in enabling verifiable claims of "green" hydrogen, combating greenwashing, and facilitating market access by providing auditable proof of low-carbon credentials to buyers, regulators, and investors [77]. However, even when only tracking and provenance data are recorded, small datasets can indirectly reveal sensitive information such as supplier relationships, production volumes, logistics patterns, and operational timelines. When combined with external data sources, this metadata could expose strategic business insights. Blockchain's immutability and transparency can increase these risks, as recorded information cannot be easily changed or removed. Even with permissioned blockchains, insider access or cross-participant data correlation remains a concern. To address these risks, approaches such as data minimization, off-chain storage, selective disclosure, and privacy-preserving cryptographic

techniques are needed to balance transparency with corporate confidentiality. Ultimately, the success of blockchain-based Digital Product Passports depends on designing systems that provide verifiable accountability while protecting enterprises' legitimate commercial interests.

4) UNIVERSAL GLOBAL CARBON EMISSION TRADING

UG-CET is a multi-blockchain architecture designed to create a decentralized, interoperable global marketplace for carbon emissions allowances and credits, enabling nations, corporations, and regional trading schemes to transact in real time across borders with transparent, auditable records [26]. Built on interconnected blockchain networks, potentially combining public chains for openness with private chains for confidentiality, it aims to overcome fragmentation in current carbon markets by standardizing verification, eliminating intermediaries, and enabling participation from developing economies previously excluded due to infrastructure or trust barriers [26]. Its strength lies in its ambition: by leveraging smart contracts for automated compliance, tokenized emission rights, and immutable ledgers, UG-CET enhances market efficiency, reduces fraud, and promotes equitable global climate accountability [26].

5) ZERO-KNOWLEDGE PROOF PROTOCOL

zk-SNARK for carbon reporting is a cutting-edge privacy-preserving blockchain technology that enables companies to cryptographically prove their compliance with carbon emission targets or reporting standards, such as Scope 1–3 disclosures, without ever exposing the underlying raw data [55]. Leveraging zk-SNARKs, the system allows an organization to demonstrate, for example, that its total emissions fall below a regulatory threshold, or that its offsets are valid and verified, while keeping all operational details, such as energy consumption figures, supplier lists, production volumes, or facility locations, completely confidential [55]. This breakthrough addresses the core tension in corporate sustainability: the need for external trust versus internal confidentiality. Its primary strength lies in its elegant cryptographic design: verifiers (regulators, investors, auditors) can be mathematically assured of accuracy without needing to see the data itself, thereby eliminating risks of competitive exposure, supply chain leaks, or data breaches [55].

6) DP-FedSW Framework

The DP-FedSW Framework is an AI-driven, federated learning system designed for industrial carbon monitoring, enabling factories, plants, and manufacturing networks to collaboratively train predictive models for emissions without pooling raw operational data in a central server [82]. By leveraging differential privacy, a mathematical technique that adds calibrated noise to outputs, it ensures that individual facility data remains locally stored and protected, while still contributing to a globally accurate model of energy use, process inefficiencies, and emission trends across the network [82]. This architecture eliminates the risks

associated with centralized data lakes: no sensitive metrics, such as production volumes, equipment specifications, fuel mixtures, or shift schedules, are transmitted or exposed. Instead, only encrypted model updates (e.g., gradients or parameter adjustments) are shared, which are further obfuscated by differential privacy guarantees to prevent inference attacks or re-identification [82], [89]. Its key strength lies in its dual innovation: combining federated learning's decentralized structure with AI-powered predictive analytics to deliver real-time, adaptive emissions forecasting tailored to complex industrial environments, ideal for smart factories and Industry 4.0 ecosystems [82].

7) SMART GRID AI SYSTEMS

Smart Grid AI Systems represent a class of advanced ML and artificial intelligence platforms integrated into modern electrical grids to dynamically optimize energy distribution, forecast demand, balance intermittent renewable sources, and minimize overall carbon emissions [75]. By analyzing real-time data from smart meters, transformers, distributed generation (e.g., rooftop solar), and consumer devices, these systems enable utilities to reduce waste, prevent outages, and accelerate the transition to clean energy, at scale [75]. Their strength lies in predictive analytics: AI models can anticipate peak loads, adjust pricing signals in real time, reroute power based on weather and usage patterns, and even identify inefficient appliances or buildings, all contributing to significant, system-wide emission reductions [75].

E. EXPERIMENTAL AND RESEARCH-ORIENTED TOOLS

This group includes Carbon Tracker, Green Algorithms, ML CO2 Impact, Experiment Impact Tracker, Cumulator, CO2web Observatory, COMPAIR Dashboard, Dreijerink's digital services model, among others. Academic research, computational impact assessments, awareness campaigns, and methodological innovation. They are designed for experimentation and transparency in emissions measurement. Widely used in academic work on the carbon footprint of computation (e.g., AI training). Public-facing dashboards such as CO2web and COMPAIR also combine awareness with research. However, often limited in scope, focusing on specialized areas (e.g., ML training, digital services). Tools may lack user-friendliness for non-experts. Most research tools are neutral in terms of privacy because they utilize system-level or aggregated data, rather than personal information. However, user-facing dashboards that require inputs (e.g., COMPAIR) still involve privacy considerations. Table 6 summarizes each of these tools.

1) CARBON TRACKER

Carbon Tracker is a lightweight, open-source Python package designed to monitor and estimate the energy consumption and associated carbon emissions of computational experiments, particularly AI and ML training jobs, by capturing real-time hardware metrics such as CPU/GPU utilization, power draw, and runtime [49]. It integrates seamlessly into scientific

TABLE 6. Experimental and research-oriented tools.

Tool	Applications	Strengths
Carbon Tracker [49]	Research, AI/ML	Research integration
Green Algorithms [49]	Research awareness	Simple, accessible
Experiment impact tracker [49]	AI/ML sustainability	Precise runtime data
ML CO2 impact [49]	AI/ML research	Geographic sensitivity
Cumulator [49]	Distributed ML research	Covers networking impact
CO2web observatory [50]	Awareness, comparisons	Product + personal insights
COMPAIR dashboard [41]	Public policy, education	Stakeholder engagement
Digital services carbon models [69]	ICT sustainability research	Expands scope to digital

workflows (e.g., Jupyter notebooks, PyTorch/TensorFlow scripts) without requiring changes to code structure, making it an indispensable tool for researchers aiming to quantify and reduce the environmental footprint of their computational work [49]. Its strength lies in its simplicity, reproducibility, and academic adoption. By providing transparent, per-experiment emission estimates tied to local grid intensity data, it fosters accountability and encourages energy-efficient algorithm design within the research community [49].

2) GREEN ALGORITHMS

Green Algorithms is a simple, web-based calculator designed to raise awareness about the carbon footprint of computational tasks, particularly in academic and AI/ML research, by estimating emissions based on user-provided inputs such as hardware type (CPU/GPU), runtime duration, cloud provider, and geographic location [49]. Unlike automated tracking tools that monitor system activity in real-time, Green Algorithms relies entirely on the manual entry of technical parameters, requiring users to input only high-level details about their computing environment [49]. This deliberate simplicity makes it highly accessible to researchers, students, and developers without requiring installation, system access, or data permissions. Its primary strength lies in its educational mission: by translating abstract metrics, such as “hours of GPU usage,” into tangible CO2 equivalents, it fosters mindfulness around energy-intensive computation and encourages more sustainable practices, including choosing efficient algorithms, optimizing code, or selecting green data centers [49].

3) EXPERIMENT IMPACT TRACKER

Experiment Impact Tracker is an open-source Python library designed to precisely monitor hardware utilization, such as CPU, GPU, and memory power consumption, in real time during computational experiments, particularly in ML and

scientific computing [49]. By integrating directly into Python workflows (e.g., Jupyter notebooks or training scripts), it captures granular system metrics during execution and estimates associated carbon emissions using regional grid intensity data, offering researchers an accurate, audit-ready measure of their computational footprint [49]. Its strength lies in its precision and seamless compatibility with existing research pipelines: unlike manual calculators, it automatically logs runtime energy use without requiring users to estimate parameters, making sustainability tracking a natural part of the experimental process [49].

4) ML CO2 Impact

ML CO2 Impact is a straightforward, web-based calculator designed to raise awareness of the carbon footprint associated with training and deploying ML models [49]. Users manually input key technical parameters, such as training duration, hardware type (e.g., GPU model), number of GPUs used, and the geographic location of the data center, to generate an estimated CO2 emission value based on region-specific electricity grid carbon intensities [49]. Its primary strength lies in its clarity and educational focus: by highlighting how emissions vary dramatically depending on location (e.g., running a job in Iceland (mostly renewable) versus Poland (coal-heavy)), it makes visible the often-overlooked environmental cost of AI and encourages more sustainable choices, such as shifting workloads to greener regions or optimizing model efficiency [49].

5) CUMULATOR

Cumulator is a Python package specifically designed to estimate the energy consumption of ML models, with a unique focus on the often-overlooked impacts of distributed training and network communication, such as data transfer between nodes, synchronization overhead, and inter-node latency [49]. Unlike general-purpose tools that track only hardware usage, Cumulator models the full lifecycle of computation in multi-device environments, quantifying how communication patterns in federated or parallelized training contribute to overall carbon emissions [49]. This makes it especially valuable for researchers working on large-scale AI systems, where network traffic can account for a significant portion of total energy use [49]. Its strength lies in its technical precision and integration into scientific workflows, as it operates within existing ML frameworks (e.g., PyTorch, TensorFlow) and provides granular, code-level insights without requiring changes to the model architecture [49].

6) CO2web Observatory

The CO2web Observatory is a web-based platform that uniquely bridges personal carbon footprinting with a comprehensive, open-access database of product-level emissions, enabling users to explore both their individual lifestyle impact and the environmental costs of specific goods, ranging from electronics and clothing to food and transportation [50]. By linking everyday purchases to life-cycle

assessment data, it empowers consumers and educators to make informed, low-carbon choices through direct comparisons (e.g., “This shirt emits 20 kg CO₂e, how does it compare to alternatives?”) [50]. Its strength lies in this dual perspective: it doesn’t just ask what you do, but what you buy, transforming abstract climate goals into tangible consumer decisions [50].

7) COMPAIR Dashboard

COMPAIR Dashboard is an interactive simulation platform designed to engage citizens, urban planners, and policymakers in participatory climate decision-making by modeling the environmental and social impacts of policy scenarios, such as low-emission zones, public transit expansion, building retrofits, or renewable energy adoption [41]. It combines intuitive calculators, scenario sliders, and visualizations to translate complex emissions data into accessible, real-time outcomes, enabling users to explore “what-if” futures at the neighborhood or city level [41]. Its key strength lies in bridging public awareness with policy design: it empowers non-experts to contribute meaningfully to sustainability planning while providing officials with evidence-based feedback loops that reflect community preferences [41].

8) DIGITAL SERVICES CARBON MODELS

Digital Services Carbon Models are methodological frameworks designed to estimate the carbon footprint of digital services, such as video streaming, cloud computing, data center operations, and app usage. This approach moves beyond traditional device-level assessments to capture the systemic energy demands of intangible digital infrastructure [69]. These models rely on aggregated, service-level parameters, such as data transfer volumes, server utilization rates, network transmission distances, and regional grid carbon intensities, rather than individual user behavior or personal identifiers [69]. By applying life-cycle analysis principles to digital workflows, researchers and tech firms can evaluate the environmental costs of services at scale, informing sustainable design choices in software architecture, content delivery networks, and energy procurement strategies [69]. Their strength lies in expanding carbon accounting into previously overlooked domains: while most tools focus on physical consumption, these models reveal that a single hour of HD streaming or a cloud-based AI query carries a measurable, and often underestimated, carbon burden [69].

V. EVALUATION

The evaluation of carbon footprint tools in this study was guided by five interrelated dimensions of privacy risk, adapted from established data protection frameworks such as ISO/IEC 29100 [21] and the General Data Protection Regulation (GDPR) [20]. These dimensions are the type of data collected, data granularity, storage and processing methods, and potential for secondary use or misuse. The five evaluation dimensions align closely with core GDPR principles, including lawfulness and purpose limitation, data minimization,

TABLE 7. Privacy risk evaluation framework for carbon footprint tools.

Dimension	Low risk	Medium risk	High risk
Type of data collected [21, 62]	Aggregate, non-personal, or technical data (e.g., server runtime, material specs, LCA databases)	Lifestyle or organizational activity (e.g., travel habits, energy bills, procurement logs)	Sensitive personal/corporate data (e.g., financial transactions, GPS, trade secrets, health proxies)
Granularity of data [63, 65]	Aggregate/annual levels (e.g., total kWh/year, sector averages)	Semi-detailed (e.g., daily usage, household-level consumption)	Real-time, fine-grained (e.g., appliance-level smart meters, continuous GPS, transaction-level blockchain logs)
Data storage & processing [20, 64]	Local processing, no storage, or anonymized aggregation (e.g., offline calculators, federated learning)	Centralized cloud storage with limited sensitivity, optional user control	Centralized, identity-linked, indefinitely stored, shared across platforms (e.g., bank-integrated apps, IoT dashboards)
Potential for secondary use / misuse [64-66]	Academic/research-only use, no commercial intent	Data may be used for advertising, benchmarking, or competitive analysis	Profiling, insurance underwriting, surveillance, market manipulation, or re-identification
User agency and control [20, 21]	Granular consent options; data access and portability; enforceable deletion; transparent privacy dashboards	Partial consent mechanisms; limited data access; unclear retention policies	Opaque or blanket consent; no deletion or portability; unclear or hidden data usage practices

accuracy, storage limitation, and integrity and confidentiality [20]. Accordingly, the tool-specific analyses that follow interpret privacy risk not only in technical terms, but also in relation to established regulatory and ethical expectations for responsible data processing [20], [21]. Table 7 summarizes these dimensions.

Low, Medium, and High privacy risk levels were assigned using rule-based criteria derived from the five evaluation dimensions, with any dimension indicating continuous, sensitive, or identity-linked data collection elevating the tool to High risk. The first dimension, type of data collected, recognizes that not all data carry equal sensitivity. Tools that rely on aggregate, non-personal, or technical system information, such as server runtime or material specifications, are considered low risk [62]. In contrast, those that gather lifestyle or organizational activity data, such as household energy bills or operational reports, are classified as medium risk. The highest risks arise when tools handle sensitive personal or corporate information, including GPS locations, financial transactions, or trade secrets [65].

The second dimension, data granularity, assesses the level of detail in the information [63]. Tools that utilize aggregated data, such as annual electricity use or sectoral averages, remain at low risk. Semi-detailed data, such as household-level consumption or daily behavior logs, present a medium risk [63]. By contrast, real-time, highly detailed tracking, such as appliance-level smart meters, continuous GPS tracking, or blockchain-based transaction data, represents a high risk due to its potential to reconstruct individual or organizational profiles [21]. The third-dimension concerns storage and processing methods. Privacy risks remain minimal when data are anonymized, processed locally, or not stored, as seen in manual-entry calculators or federated learning approaches [64]. A medium risk is associated with data that are centrally stored on third-party servers but lack high sensitivity. The highest risk emerges when data are centralized, identity-linked, and potentially shared across platforms or stored indefinitely, a practice common in cloud-based IoT or finance-integrated tools [20].

Fourth, the potential for secondary use or misuse reflects the likelihood that data may be repurposed beyond their original intent [64]. Academic research tools and offline calculators, with limited scope for repurposing, pose a low risk. Lifestyle apps or organizational reporting platforms that could be shared with advertisers or competitors are medium risk [66]. High-risk scenarios involve tools that collect financial data, IoT household monitoring, or transparent blockchain records, where the possibility of misuse is significant [65]. By synthesizing these five dimensions, we assign an overall privacy impact level, low, medium, or high, to each tool. All five privacy dimensions were initially treated as equally weighted to ensure transparency, comparability across heterogeneous tool categories, and to avoid introducing subjective bias in the absence of validated sector-specific weighting standards in the privacy literature [97], [98]. Tools are considered low risk if they collect non-sensitive, aggregated data with minimal re-identification potential; medium risk if they handle moderately sensitive or semi-detailed information; and high risk if they involve sensitive, granular, or identity-linked data with significant potential for misuse. This framework ensures a consistent and transparent evaluation across all categories of tools.

Finally, the fifth dimension, user agency and control, captures the extent to which users (individuals or organizations) can meaningfully understand, manage, and exercise rights over their data [21]. This dimension reflects GDPR's privacy by design and by default principle (Article 25) and recognizes that tools with similar data collection practices may differ substantially in privacy impact depending on governance mechanisms [20]. Tools that provide granular consent options, data access and portability, enforceable deletion or retention limits, and transparent privacy dashboards are assessed as lower risk [21]. Conversely, tools that rely on opaque, take-it-or-leave-it terms, lack deletion mechanisms, or obscure data usage practices present

higher privacy risks, even when the underlying data appear comparable [20].

Explicit criteria were employed in classifying the Low, Medium, or High privacy risk tools to upgrade the transparency and reproducibility. Each carbon footprint tool was judged against the five dimensions given in Table 7: data type, data granularity, data storage and processing, and potential for secondary use. When tools relied on non-personal or aggregated data, used coarse measurements, and applied local or anonymized processing with no clear secondary use, they were classified as Low risk [20]. Medium risk tools had features such as moderately sensitive data, semi-detailed information, or centralized storage with limited exposure [20]. High-risk tools obtained sensitive personal or corporate data, employed fine-grained or real-time data, depended on centralized and identifiable storage, or had clear potential for secondary use [20]. In borderline instances, a conservative approach was taken; tools that exhibited high-risk characteristics in more than one aspect, or that combined sensitive data with high granularity or centralized storage, were moved to the higher risk category.

While this study does not conduct legal compliance audits, the evaluation framework is explicitly grounded in GDPR and ISO/IEC 29100 principles, allowing regulatory responsibilities to be interpreted at a categorical level. Individual and household carbon calculators typically function as data controllers under GDPR, with primary obligations related to lawful consent, purpose limitation, and data minimization [68], [85]. Corporate and organizational platforms often operate in joint controller or controller-processor arrangements, where responsibilities extend to safeguarding commercially sensitive data, ensuring integrity and confidentiality, and managing secondary use across reporting, auditing, and benchmarking contexts [41], [71]. Infrastructure, building, and city-level tools generally implicate lower regulatory exposure, as they rely predominantly on aggregated or technical datasets; however, tools incorporating occupancy or household-level inputs may trigger additional obligations related to proportionality and transparency [49], [55]. IoT- and blockchain-based systems pose the strongest regulatory tensions, as continuous sensing and immutable records challenge storage limitation, controllability, and the right to erasure [50], [70]. By contrast, research-oriented and privacy-preserving frameworks (e.g., federated learning or zero-knowledge proofs) align closely with GDPR's privacy-by-design principle by minimizing disclosure while enabling verification. This category-level regulatory mapping strengthens the transparency and reproducibility of the evaluation without implying formal compliance judgments [70].

A. INDIVIDUAL AND HOUSEHOLD CARBON CALCULATORS

From a GDPR perspective, most individual carbon calculators align with the principles of data minimization and purpose limitation, as they rely on self-reported, coarse-grained

TABLE 8. Privacy evaluation of individual carbon footprint tools mapped against GDPR principles.

Tool	Privacy level	Reason
WWF Calculator [68, 85]	Low	Manual, non-tracking questionnaire
CoolClimate [67, 85]	Medium	Detailed manual inputs may form sensitive profiles
MyClimate [72]	Low	Minimal voluntary inputs, nonprofit, limited retention
REAP Petite [61]	High	Deep consumption-based inputs, sensitive & granular
CO2web Observatory [50]	Medium	Product DB neutral, but personal calculator data may be stored
Doconomy / Joro / Svalna [52, 53]	High	Bank/transaction linking, highly sensitive financial data
EcoTrack [76]	High	Detailed lifestyle data, stored in third-party cloud
ECOSAUR [81]	High	Real-time data streaming, continuous behavioral profiling risk
Tanaffas [78]	High	GPS, shopping, AR tracking, persistent device-level identifiers

inputs and do not require continuous monitoring or long-term storage [20]. However, tools that integrate financial or mobility data may raise concerns regarding lawfulness and transparency, particularly if consent mechanisms and data retention policies are not clearly specified [20]. Most individual and household carbon calculators differ significantly in their impact on privacy, depending on their data collection mechanisms. Simple tools, such as the WWF Calculator and MyClimate, are low risk, as they rely on voluntary, aggregate inputs without persistent storage. Tools like CoolClimate and CO2web Observatory fall into the medium-risk range, as their detailed or stored inputs can form sensitive profiles. By contrast, consumption-based calculators like REAP Petite, finance-linked apps such as Doconomy and Joro, and real-time or GPS-enabled platforms like EcoTrack, ECOSAUR, and Tanaffas are high risk, given their granular, sensitive, and continuous data collection.

Beyond data type and granularity, privacy risk in individual and household carbon calculators is also shaped by the degree of user agency afforded by each tool. In line with GDPR principles of transparency, consent, and user rights (Articles 7, 12–15, 17, and 25), tools that allow users to control data submission, avoid account creation, and delete or withhold data exhibit substantially lower privacy impact than tools with opaque, take-it-or-leave-it consent models [20]. Accordingly, two tools with similar data collection practices may differ significantly in privacy risk depending on whether users can meaningfully exercise control over how their data are processed, retained, or withdrawn. Table 8 summarizes the privacy level of each tool in this category.

1) WWF Footprint Calculator

The WWF Footprint Calculator operates at a low privacy level due to its voluntary, manual questionnaire design, which collects only aggregated, self-reported data on housing, travel, and diet, without the use of sensors, GPS, or third-party integrations [85]. It avoids sensitive fields like income, health, or financial records, and does not track users continuously or link to external databases. Data is typically stored locally or in minimal server logs, with no persistent profiling; secondary use for marketing or research is rare and is disclosed transparently [68]. Its simplicity ensures users retain complete control: inputs are ephemeral, non-identifiable, and never monetized. With no behavioral tracking or inference engines, the risk of re-identification or exploitation is negligible. Its strength lies in ethical minimalism, promoting awareness through transparency rather than surveillance. By design, it empowers rather than exploits, making it a benchmark for privacy-conscious environmental tools [85]. From a user agency perspective, the WWF Footprint Calculator fully aligns with GDPR's privacy-by-design principle, as it requires no account creation, offers implicit session-based consent, and provides users complete control over whether data are entered at all [68].

2) CoolClimate Calculator

The CoolClimate Calculator poses a medium privacy risk due to its detailed, voluntary collection of lifestyle and operational data, such as utility bills, vehicle mileage, and spending habits. Although this data is not automatically linked to bank or smart meter accounts, it can still be used to construct highly sensitive behavioral profiles when aggregated [85]. Even without passive tracking, the granularity of responses enables inferences about income, health, mobility, and consumption ethics, creating a potential risk of re-identification or profiling if stored on servers. Data is often retained for user profiles or longitudinal analysis, increasing exposure to breaches or unintended reuse [67]. While not inherently commercialized, secondary uses, like research aggregation or targeted nudges, may occur without explicit consent. Its scientific value comes at an ethical cost: the more precise the footprint estimate, the more it reveals about private life [67]. Unlike minimal tools, it trades transparency for depth, making user trust a critical factor. Without strong encryption, anonymization, or clear data retention policies, this medium-risk profile demands greater governance to ensure sustainability doesn't become surveillance [85]. However, user agency mechanisms remain limited: consent is largely implicit, data retention policies are not always prominently disclosed, and users have constrained ability to explicitly delete historical inputs once stored [67].

3) MyClimate

MyClimate operates at a low privacy level due to its minimalist, nonprofit design, collecting only basic, low-sensitivity lifestyle inputs, such as travel distance or diet, without

requiring personal identifiers, location data, or financial details [72]. Data is rarely stored beyond the session, unless users opt in to register carbon offsets. Even then, retention is purpose-limited and transparent. As a nonprofit, it has no incentive for ad monetization, behavioral profiling, or sharing third-party data. Storage is minimal, with no persistent user profiles or analytics tracking by default [72]. The secondary use of data for research is rare and typically anonymized, with explicit consent being sought where applicable. MyClimate prioritizes ethical integrity over data extraction, offering credible emissions estimates while safeguarding user autonomy [72]. Its strength lies in restraint: it asks just enough to inform, not enough to infer. In a landscape crowded with surveillance-driven sustainability tools, MyClimate stands as a model of trust-based design, where climate action and digital privacy are not competing goals, but aligned values [72]. Importantly, MyClimate provides a high degree of user agency by avoiding mandatory registration, clearly distinguishing optional offset-related data from calculation inputs, and allowing users to disengage without data persistence [72].

4) REAP Petite

REAP Petite carries a high privacy risk due to its consumption-based design, which demands highly granular inputs, such as shopping receipts, brand-specific spending, and detailed lifestyle habits, that serve as proxies for income, health, ethics, and socioeconomic status [61]. This depth enables precise emissions modeling but also creates rich behavioral profiles that are vulnerable to re-identification, inference, and profiling, even when the data is anonymized. Data is typically stored persistently to track trends and support longitudinal analysis, increasing exposure to breaches or unauthorized access [61]. Its academic rigor comes at the cost of intrusion: spending patterns can reveal intimate details, such as dietary restrictions, medical needs, or political affiliations. Secondary use for research may be well-intentioned, but without stringent governance, it risks enabling commercial exploitation or discriminatory targeting. REAP Petite exemplifies the ethical tension in advanced carbon accounting: its power lies in detail, but that detail erodes privacy. To remain legitimate, it must enforce strict data minimization, end-to-end encryption, user-controlled deletion, and transparent consent, not only for accuracy but also for dignity [61]. From a user agency standpoint, REAP Petite exhibits elevated risk: while participation is voluntary, users have limited practical control over downstream data use once detailed consumption data are submitted for longitudinal analysis [61].

5) CO2web Observatory

CO2web Observatory presents a medium privacy risk due to its hybrid model: while its product emissions database is neutral and public, the optional personal calculator invites detailed lifestyle inputs, such as travel, diet, and spending, which, when stored or linked to profiles, can generate behavioral signatures revealing income, habits, or values [50].

Although CO2WEB is not involved in the collection of transaction-level data, the repeated inputs that are stored for trend analysis can still unveil the highly sensitive lifestyle patterns of the users [50]. These patterns may include dietary preferences, housing characteristics, mobility behavior, or consumption intensity, which may indirectly indicate income level, health-related choices, or environmental values [67]. If these patterns are linked with demographic context, such as location, household size, or age group, the continuous patterns significantly reduce the level of anonymity and thus increase the possibility of profiling or identification even if there are no direct personal identifiers. Granularity is moderate, not as invasive as receipt-based tools, but sufficient to infer sensitive patterns [50]. Storage practices vary by implementation, often relying on server-side retention without explicit opt-out defaults. While no third-party tracking or ad monetization occurs, secondary use in research or policy may blur the boundaries of consent [50]. Its strength, connecting individual choices to product impacts, also creates ethical ambiguity: awareness shouldn't require surrendering privacy. To mitigate risk, clear data retention policies, local processing options, and granular consent controls are essential to ensure transparency doesn't become exploitation [50]. Although users may choose whether to save their data, consent mechanisms are often broad rather than granular, and explicit deletion or portability options are inconsistently documented [50].

6) DOCONOMY, JORO, AND SVALNA

Doconomy, Joro, and Svalna pose a very high privacy risk by directly linking to bank and credit card accounts, harvesting transaction-level data that reveals intimate details, such as dietary choices (organic vs. processed), medical purchases, travel patterns, donations, and even social habits, enabling precise behavioral profiling with trivial re-identification potential [53]. This data is centralized on corporate servers, often without transparent retention policies or user-controlled deletion, creating prime targets for breaches or misuse [52]. The granularity enables the inference of income, health status, political leanings, and lifestyle vulnerabilities, making it highly attractive to advertisers, insurers, and even law enforcement [52]. While framed as sustainability tools, their business models inherently monetize behavior, turning climate awareness into surveillance capitalism. These platforms exploit the trust placed in environmental advocacy to normalize invasive data collection [53]. Their "convenience" comes at the cost of autonomy: users trade the most sensitive aspects of their financial lives for carbon estimates. Critically, user agency is severely constrained in these platforms: consent is typically bundled, data portability is limited, and meaningful deletion of transaction histories is often unclear or technically infeasible once financial integrations are established [52]. This lack of effective control, despite GDPR's emphasis on user rights, significantly amplifies the already high privacy risk posed by transaction-level carbon accounting.

7) EcoTrack

EcoTrack carries a high privacy risk due to its reliance on detailed lifestyle inputs, including travel, diet, and energy use, which are processed via a centralized cloud infrastructure (Google Cloud), placing sensitive user data under third-party control [76]. While inputs are manually entered, their combination with persistent cloud storage enables behavioral profiling, pattern inference, and potential re-identification, even if anonymized. The use of a commercial platform introduces risks of undisclosed data sharing, algorithmic bias, or secondary use for advertising, research, or product development without explicit consent [76]. Though granularity is moderate, the aggregation of habits across domains (e.g., frequent organic purchases + electric vehicle use) can reveal socioeconomic status, ethics, or health conditions. Storage is not local, which limits user sovereignty over data retention and deletion. EcoTrack exemplifies the growing trend where sustainability tools outsource trust to corporate cloud providers, trading privacy for convenience [76]. Despite the implementation of technical and organizational safeguards, data collection and storage by cloud service providers pose potential threats to individual privacy. The centralization of data diminishes user control over retention, deletion, and secondary utilization, as the providers govern these aspects. Cloud infrastructure facilitates profiling and inference beyond its original purposes; for instance, aggregated data, metadata such as access logs, and usage patterns may reveal sensitive insights. This shift in trust from users to providers establishes dependencies that can ultimately weaken privacy protections, even with existing safeguards in place. While EcoTrack allows users to manually enter data, user agency is weakened by reliance on centralized cloud storage and limited transparency regarding retention duration and secondary processing [76]. The absence of robust privacy dashboards or guaranteed deletion mechanisms elevates governance-related risk, even when collection is not fully automated.

8) ECOSAUR

ECOSAUR presents a high privacy risk due to its real-time, streaming architecture, which utilizes Kafka and Spark to continuously ingest sensor and usage data, capturing granular, persistent behavioral traces such as location, mobility patterns, consumption timing, and activity frequency [81]. This constant monitoring enables detailed profiling: even anonymized streams can be re-identified through unique behavioral signatures, revealing intimate habits such as work schedules, social routines, or health-related behaviors. Data is likely stored centrally for analytics and personalization, increasing exposure to breaches, insider access, or undisclosed secondary uses, such as behavioral targeting or third-party sharing [81]. Unlike tools relying on sporadic inputs, ECOSAUR's design normalizes surveillance by treating lifestyle as a live feed. This "always-on" model transforms sustainability into a form of digital oversight, where environmental insight comes at the cost of continuous

observation [81]. User agency is particularly limited in ECOSAUR's streaming architecture, as continuous data ingestion constrains meaningful consent withdrawal once monitoring begins [81]. The lack of clear pause, deletion, or data minimization controls places ECOSAUR at the highest end of the privacy risk spectrum when governance and control are considered alongside technical design.

9) TANAFFAS APP

Tanaffas poses a high privacy risk due to its invasive, gamified design, as it continuously collects GPS location logs, real-time shopping receipts, device identifiers, and camera access for AR product scanning, all of which are fused into centralized behavioral profiles [78]. This granularity reveals not only consumption patterns but also health habits, religious practices, political affiliations (e.g., donations), social circles, and daily routines, all with high re-identification potential. Gamification incentivizes sharing achievements and comparisons, further exposing data through social networks and increasing the risk of profiling or misuse. Centralized storage of sensitive, persistent data creates attractive targets for breaches or unauthorized access by third parties [78]. Tanaffas weaponizes engagement, turning climate action into a surveillance-driven performance where users trade intimate life details for badges and leaderboards. The tool's power lies in its depth, but so does its danger: by normalizing constant monitoring under the banner of sustainability, it erodes digital autonomy [78]. Moreover, Tanaffas offers minimal user agency: consent is embedded within gamified participation, data sharing is incentivized rather than optional, and effective data deletion or scope limitation is poorly defined [78]. This asymmetry between user engagement and control exemplifies how persuasive design can undermine informed consent, intensifying privacy risk beyond data collection alone.

B. CORPORATE AND ORGANIZATIONAL TOOLS

Corporate carbon platforms introduce more substantial privacy risks in light of GDPR principles. Their integration with ERP and financial systems challenges purpose limitation and storage limitation, as emissions data may be retained indefinitely and reused for compliance, auditing, or benchmarking [20]. Moreover, the aggregation of operational and supplier data increases the importance of integrity and confidentiality safeguards as required under GDPR and ISO/IEC 29100. Corporate and organizational carbon tools exhibit varying privacy impacts, depending on how deeply they integrate with company systems. Privacy-by-design tools, such as GES 1point5, along with research-oriented utilities like CodeCarbon, EmissionsTracker, Green Metrics Tool, and FootPrinter, pose a low risk as they rely on aggregated or technical data with minimal identifiers. Methodologies such as the Greenhouse Gas Protocol, Bilan Carbone, Carbon Footprint Ltd., and the Carbon Trust fall into the medium-risk range due to their reliance on detailed operational inputs, often processed by third parties. The highest risk emerges with Carbon Analytics Dashboards, which directly

TABLE 9. Privacy evaluation of corporate and organizational carbon tools mapped against GDPR principles.

Tool	Privacy level	Reason	Confidence level	Auditability / evidence basis
Microsoft cloud for sustainability [92]	High	Ingests ERP/finance/operations and supply-chain data; automated connectors and cloud storage increase granularity & secondary-use risk.	Moderate	Proprietary platform; assessment based on vendor documentation and regulatory disclosures
Salesforce net zero cloud [93]	High	Deep CRM and supplier integration (transactional and partner data); centralized cloud reporting; potential exposure of supplier/contract detail.	Moderate	Proprietary; documentation n-based assessment
SAP sustainability control tower [94]	High	Uses actuals from multiple SAP systems (ERP/SCM); high granularity and business-sensitive datasets; centralized storage.	Moderate	Proprietary; no public technical audit access
IBM environmental intelligence suite / Envizi [95]	High	Integrate operational telemetry, supply-chain and geospatial risk data; enterprise-scale centralized processing; sensitive operational exposures.	Moderate	Proprietary; assessed via white papers and technical briefs
Oracle fusion cloud sustainability / OCI carbon emissions analysis [96]	High (medium-high where only OCI workload estimates used)	Oracle's Fusion/EPM and OCI Carbon features ingest ERP/SCM & cloud workload telemetry; likely centralized. OCI enterprise Fusion integration drives high risk.	Moderate	Proprietary; documentation n-based
Google Carbon API [44, 87]	Medium	Requires workload logs; stored	Moderate	Proprietary API; limited transparency

access sensitive financial and operational systems, creating significant exposure.

TABLE 9. (Continued.) Privacy evaluation of corporate and organizational carbon tools mapped against GDPR principles.

		by cloud vendor		on retention and reuse
GES 1point5 [45]	Low	Anonymization, GDPR alignment, minimal identifiable data	High	Peer-reviewed, research-audited implementation
Greenhouse Gas Protocol / Bilan Carbone [45, 60]	Medium	Requires detailed organizational data, depends on implementation	High	International standards; transparent methodology
Carbon Footprint Ltd. [67]	Medium	Third-party hosted, retention and usage risks	Moderate	Commercial provider; limited technical disclosure
Carbon Trust Tools [67]	Medium	Sensitive operational data, handled by external providers	Moderate	Institutional trust-based; not open-source
Carbon Analytics Dashboards [41]	High	Deep integration with finance/ops; exposure of competitive and employee data	Low–Moderate	Vendor-specific, opaque implementations
CodeCarbon [88]	Low	Local telemetry only, no personal data	High	Open source; fully auditable
EmissionsTracker [88]	Low	Local estimates, no personal identifiers	High	Open-source and research-oriented, auditable implementation
Green Metrics Tool [44]	Low	Isolated monitoring, system-level only	High	Open source; auditable
FootPrinter [79]	Low	Simulation-based, no individual data	High	Research-based; no live data ingestion

In corporate and organizational contexts, privacy risk is further shaped by the degree of organizational user agency, defined here as the ability of firms to control data flows, consent scopes, retention periods, portability, and deletion across integrated systems. While GDPR formally applies to personal data, its principles of transparency, purpose limitation, and privacy by design (Articles 5 and 25) remain highly relevant to corporate sustainability data governance. Tools that limit organizational visibility into how data are processed, reused, or retained by vendors significantly reduce practical control, even where contractual safeguards exist. Consequently, enterprise platforms with similar data access profiles may differ substantially in privacy impact depending on whether organizations retain sovereignty over their emissions data or effectively delegate it to third-party infrastructures. Table 9 summarizes the privacy level of each tool in this category.

The confidence levels provided in Table 9, are defined as follows:

- High confidence: Tool architecture, data flows, and privacy properties are verifiable through open-source code, peer-reviewed publications, or auditable technical documentation.
- Moderate confidence: Assessment is based on detailed vendor documentation, white papers, regulatory filings, or reputable secondary analyses, but without direct audit access.
- Low confidence: Limited public disclosure of internal data handling, storage, or secondary use practices; assessment relies primarily on marketing materials or high-level descriptions.

1) MICROSOFT CLOUD FOR SUSTAINABILITY

Microsoft Cloud for Sustainability poses a high privacy risk due to its deep integration with enterprise systems, including ERP, finance, supply chain, and Azure cloud services, which automatically ingest granular operational data such as energy use, production volumes, logistics routes, and supplier emissions. This centralized, highly sensitive business intelligence on Microsoft's cloud infrastructure creates significant exposure to breaches, insider access, or unintended secondary use, for example, profiling suppliers or inferring competitive strategies from aggregated patterns [92]. Automated connectors increase data volume without user consent oversight, and proprietary algorithms may repurpose inputs beyond sustainability goals [92]. Despite strong security protocols, the lack of local processing or actual data minimization means organizations surrender control over their most confidential environmental and operational metrics. Privacy is not just compromised, it's systemic [31]. From a user agency perspective, organizations have limited practical control once automated connectors are enabled: consent is typically broad, data flows are continuous, and granular deletion or selective withdrawal of specific datasets is constrained [92]. This asymmetry between data contribution and governance authority significantly amplifies privacy risk beyond technical exposure alone.

2) SALESFORCE NET ZERO CLOUD

Salesforce Net Zero Cloud poses a high privacy risk due to its deep integration with CRM and supplier networks, which ingests transactional data, contract terms, procurement histories, and partner emissions profiles into a centralized cloud environment [93]. While this enables automated ESG reporting and supply chain engagement, it exposes sensitive commercial relationships, such as vendor performance, pricing agreements, or sourcing strategies, that could be leaked, re-identified, or exploited by competitors or third parties. Even anonymized datasets may be reconstructed through network analysis or cross-referencing with public records. Salesforce controls data storage, processing, and access policies, leaving enterprises with limited visibility

into how their confidential sustainability data is used beyond reporting [93]. Organizational user agency is further constrained by Salesforce's platform-centric governance model, where enterprises rely on vendor-defined retention, access, and portability mechanisms. Although contractual controls exist, the lack of fine-grained, tool-level consent and deletion controls reinforces its high-risk classification under the user agency criterion.

3) SAP SUSTAINABILITY CONTROL TOWER

The SAP Sustainability Control Tower poses a high privacy risk due to its ingestion of real-time, high-granularity data from core SAP systems, including ERP and SCM, which captures sensitive operational details such as production volumes, energy consumption per facility, logistics routes, and supplier-specific emissions [94]. This enables precise, audit-ready reporting but centralizes highly confidential business intelligence within SAP's cloud infrastructure, creating concentrated exposure to breaches, insider threats, or unauthorized access by competitors. Even aggregated insights can reveal strategic patterns, such as shifts in sourcing or plant closures, that competitors may be able to infer. While role-based controls offer access management, they don't mitigate the inherent risk of storing proprietary sustainability data on a third-party platform. Enterprises gain accuracy and compliance, but at the cost of sovereignty. The tool transforms sustainability from an internal function into a centrally hosted, vendor-controlled process, where transparency for regulators comes at the expense of corporate confidentiality [94]. While SAP provides role-based access management, organizational user agency remains limited with respect to data residency, downstream reuse, and algorithmic transparency [94]. Once emissions data are embedded within SAP's cloud ecosystem, enterprises face structural barriers to enforcing selective deletion or restricting secondary analytical use, elevating governance-related privacy risk.

4) IBM ENVIRONMENTAL INTELLIGENCE SUITE/ENVIZI

IBM Environmental Intelligence Suite (with Envizi) carries a high privacy risk due to its aggregation of enterprise-scale operational telemetry, energy use, facility emissions, logistics flows, and geospatial risk data, often tied to suppliers and physical assets [95]. By centralizing this sensitive information in a cloud-based platform, it creates a high-value target for breaches or re-identification, where aggregated patterns can expose competitive strategies, supplier dependencies, or site-specific vulnerabilities. While powerful for climate risk modeling and ESG reporting, the tool's deep integrations mean enterprises surrender control over proprietary environmental data to IBM's infrastructure. Even anonymized datasets risk de-anonymization through contextual analysis. The convenience of unified analytics comes at the cost of operational confidentiality; sustainability becomes a vendor-managed function, rather than an owned asset [95]. User agency is further weakened by the platform's unified analytics model, which consolidates diverse data streams under

vendor-controlled processing pipelines. Although enterprises may define access roles, they have limited insight into how aggregated datasets may be reused or retained, reinforcing the high privacy impact from a governance standpoint.

5) ORACLE FUSION CLOUD SUSTAINABILITY

Oracle Fusion Cloud Sustainability poses a significant privacy risk when integrated with ERP, SCM, and EPM systems, as it ingests granular operational data, including production volumes, facility energy use, and supplier emissions, into a centralized Oracle cloud infrastructure [96]. This enables precise Scope 1–3 reporting but exposes sensitive business intelligence to vulnerabilities, such as breaches or competitive profiling. Even the OCI Carbon Emissions Analysis, which estimates cloud workload emissions, poses a medium-high risk if used in isolation. While aggregated cloud metrics are less intrusive, linking them to enterprise financial or supply chain systems reintroduces high-risk exposure. Centralized storage, proprietary algorithms, and limited transparency over data usage amplify concerns: organizations lose control over their environmental data, trading autonomy for automation. The tool's power lies in integration, but so does its risk [96]. From an agency perspective, Oracle's tightly coupled cloud ecosystem limits organizations' ability to isolate sustainability data from broader enterprise analytics once integration is established. Selective consent withdrawal or dataset-level deletion is technically constrained, increasing dependence on vendor trust rather than enforceable control.

6) GOOGLE CARBON API

The Google Carbon API poses a medium privacy risk because, although it handles no personal data, it requires organizations to grant access to detailed cloud usage logs, including compute instances, storage volumes, and regional resource allocation, which can reveal strategic infrastructure decisions, workload patterns, and operational priorities [44]. Although anonymized at the user level, this data may still reveal competitive insights. For instance, spikes in AI training could signal product development timelines, or shifts in geographic hosting may indicate regulatory avoidance [87]. Data resides entirely on Google's proprietary cloud infrastructure, creating vendor dependency and limited transparency around retention, access controls, or potential secondary use. While Google enforces enterprise-grade security, organizational confidentiality still hinges on trust in a single provider with conflicting commercial interests. The tool enables precision, but not sovereignty [44]. Its value lies in automation; its risk lies in centralization [87]. Although organizations voluntarily enable the API, user agency is limited to initial access provisioning, with minimal visibility into retention duration, internal reuse, or long-term storage practices.

7) GES 1point5

GES 1point5 operates at a low privacy risk due to its research-focused, privacy-by-design architecture: it collects

only aggregated, institutional-level emissions data, such as energy use or procurement volumes, without capturing personal identifiers, individual behaviors, or granular operational details [45]. All inputs are anonymized from the outset, in alignment with GDPR principles, and stored minimally with strict access controls. Outputs are designed for reporting and benchmarking, not profiling, and secondary use is intentionally limited by policy and technical safeguards [45]. No user tracking, behavioral logging, or third-party integrations occur. Its strength lies in ethical precision: it delivers scientifically rigorous carbon accounting while ensuring no individual or entity can be re-identified [45]. This makes it uniquely suitable for academic and public institutions where data sovereignty and trust are paramount. Unlike tools that trade transparency for surveillance, GES 1point5 proves that accountability need not come at the cost of privacy; it thrives when confidentiality is foundational. It sets a gold standard: sustainability tools should measure systems, not souls [45]. Importantly, GES 1point5 exhibits strong organizational user agency: data submission is deliberate, scope-limited, and transparent, with clear boundaries around storage and reuse.

8) GREENHOUSE GAS PROTOCOL / BILAN CARBONE

The Greenhouse Gas Protocol carries a medium privacy risk because, although it is a methodological framework rather than a tool, it requires organizations to input sensitive operational data across Scopes 1–3, including energy use, supply chain logistics, fleet details, and production volumes [60]. Though no personal identifiers are collected, this data can reveal competitive strategies, vendor relationships, or facility vulnerabilities, especially if stored in unsecured spreadsheets or shared internally without access controls. Granularity is moderate but can become intrusive when linked to specific sites or processes [45]. Privacy depends entirely on the implementing organization's governance: centralized storage, weak encryption, or accidental sharing can lead to leaks or re-identification through cross-referencing. These standards enable global comparability, but not protection [45]. Their strength lies in transparency for reporting; their weakness lies in assuming trust, not enforcing security [60]. User agency within these methodologies depends entirely on organizational implementation, making governance practices, such as internal access control, retention policies, and audit trails, critical determinants of privacy impact [60]. Where such controls are weak, the absence of built-in enforcement mechanisms elevates practical risk.

9) CARBON FOOTPRINT LTD. CALCULATOR

Carbon Footprint Ltd.'s Calculator carries a medium privacy risk due to its dual role as both a public tool and a commercial consultancy platform: while individual users enter basic lifestyle data, organizational clients submit sensitive operational inputs, energy use, travel logs, and supply chain volumes, hosted on centralized third-party servers [67]. Though marketed for transparency, the provider's commercial incentives raise concerns about data retention, anonymization

practices, and potential secondary use, such as aggregating insights for marketing, benchmarking services, or selling trend reports. There are no clear policies detailing how data is stored, shared, or deleted, leaving organizations vulnerable to inadvertent exposure or profiling through pattern analysis [67]. Even anonymized datasets can be re-identified when combined with public records or industry benchmarks. The tool's convenience masks structural risks: by outsourcing data to a profit-driven entity, users trade control for ease. Its value in driving awareness is real, but so is its potential for exploitation [67]. Organizational user agency is constrained by opaque policies governing data retention, reuse, and deletion, particularly for consultancy-linked services [67]. The lack of explicit, tool-level consent management mechanisms increases governance risk despite moderate data sensitivity.

10) CARBON TRUST TOOLS

Carbon Trust Tools carry a medium privacy risk because, while they are trusted by corporations and bound by confidentiality agreements, they require organizations to share sensitive operational data, such as energy consumption, manufacturing processes, and supply chain metrics, with a third-party provider [67]. Though contractual safeguards limit misuse, the fact that this data resides on external servers introduces inherent exposure: breaches, insider access, or inadvertent sharing could compromise competitive advantage or reveal proprietary workflows [67]. Granularity is often high enough to infer business strategies or facility vulnerabilities, even when data is anonymized. Secondary use for benchmarking or research typically requires consent, but opaque policies leave room for ambiguity. The tools' credibility stems from institutional trust, not technical privacy-by-design. Their strength lies in standardization and authority; their weakness lies in centralization. Organizations must assume responsibility for securing inputs, as the platform itself does not offer end-to-end encryption or local processing options [67]. While contractual trust plays a central role, user agency remains largely legal rather than technical: organizations depend on agreements rather than enforceable system controls to manage deletion, reuse, or data minimization. This reliance moderates, but does not eliminate, privacy risk.

11) CARBON ANALYTICS DASHBOARDS

Carbon Analytics Dashboards pose a high privacy risk due to their deep integration with internal corporate systems, ingesting real-time data from finance, HR, logistics, and production platforms to automate emissions reporting [41]. These grants access highly granular, sensitive operational data, including employee travel patterns, supplier contracts, manufacturing volumes, energy allocation by facility, and even procurement timelines—all of which constitute strategic competitive intelligence. Centralized storage on third-party cloud platforms increases exposure to breaches, insider threats, or unauthorized aggregation that could reveal sensitive information, such as market positioning, supply chain

dependencies, or operational inefficiencies [41]. Even anonymized outputs may be re-identified through pattern analysis or cross-referencing with public datasets. As Orfanou (2023) warns, the tool's power lies in its comprehensiveness, but so does its danger: it transforms sustainability reporting into a surveillance infrastructure for corporate operations [41]. While automation delivers efficiency, it inherently erodes confidentiality. In this model, accountability becomes vulnerability, and the line between ESG leadership and digital exposure is dangerously thin [41]. Critically, organizational user agency is weakest in this category, as real-time integrations limit meaningful consent granularity and make selective data withdrawal impractical once dashboards are operational.

12) CodeCarbon

CodeCarbon operates at a low privacy risk by design: it is an on-device Python package that measures only technical metrics, power draw, runtime, and hardware utilization, during computational tasks, with no access to personal files, user identities, network activity, or external systems [88]. All processing occurs locally, with data never uploaded, stored centrally, or shared unless explicitly exported by the user [88]. It requires no accounts, cloud dependencies, or third-party integrations, eliminating risks of profiling, re-identification, or surveillance. Its purpose is purely technical transparency, empowering researchers to quantify energy use without compromising autonomy [88]. There is no secondary use beyond user-initiated reporting, and no behavioral or organizational data is inferred or retained. In an ecosystem where sustainability tools increasingly demand access to data, CodeCarbon stands as a model of ethical minimalism, delivering accountability through measurement, not intrusion [88]. CodeCarbon offers maximal user agency: organizations retain full control over when measurements occur, where data are stored, and whether results are shared.

13) EmissionsTracker

EmissionsTracker operates at a low privacy risk by design: it estimates carbon emissions locally on the user's device using only technical inputs, hardware usage, runtime, and regional grid intensity data, without collecting personal identifiers, location history, or organizational metadata [88]. All computation occurs offline, with no cloud uploads, account creation, or third-party dependencies. Even when leveraging geolocation for grid carbon factors, it utilizes public, aggregated regional data, rather than real-time tracking or personal GPS. No behavioral patterns, system logs, or sensitive operational details are stored or transmitted [88]. Its architecture prioritizes both accuracy and autonomy: users gain precise emissions insights without surrendering control over their data. With zero profiling, minimal storage, and no secondary use, it poses a negligible risk of identification or misuse [88]. User agency is similarly strong, as all processing remains local and optional, with no mandatory reporting, storage, or transmission.

14) GREEN METRICS TOOL

The Green Metrics Tool operates at a low privacy risk by design, as it monitors only system-level metrics, power consumption, CPU/GPU load, and network traffic volume within isolated containers or execution environments, without accessing content, user identities, filenames, URLs, or behavioral data [44]. All analysis occurs locally on the user's device, with no cloud transmission, account linking, or persistent storage of inputs. Its granularity is strictly technical, capturing resource usage patterns, not who is using them or why. Green Metrics Tool enables sustainable software development through transparency, not surveillance [44]. It empowers developers to optimize energy efficiency without compromising confidentiality or exposing proprietary workflows. With no profiling, tracking, or third-party access, its minimal footprint extends beyond emissions; it extends to privacy [44]. The tool further excels in user agency by enabling fine-grained execution control and complete data isolation within development environments. Sustainability insights are generated without relinquishing governance, making it a model for privacy-preserving enterprise tooling [44].

15) FootPrinter

FootPrinter operates at a low privacy risk because it is a simulation-based tool grounded in OpenDC, using anonymized benchmark data and synthetic workload traces to model data center energy use and emissions, without accessing real-world user, organizational, or operational data [79]. It relies on aggregated, publicly available or research-generated inputs, such as server utilization patterns or cooling efficiency curves, none of which originate from live systems or identifiable sources. No personal, financial, or institutional information is collected, stored, or transmitted. Its entire purpose is predictive modeling, not monitoring, making it inherently privacy-preserving by design [79]. Even when configured with custom parameters, these remain abstracted technical inputs, not sensitive operational records. There is no user tracking, no cloud dependency, and no risk of re-identification. FootPrinter demonstrates that high-fidelity environmental analysis can be achieved without compromising confidentiality, as it measures the footprint of systems, not individuals [79]. FootPrinter provides full organizational agency by design: users define parameters without contributing live or historical operational data, ensuring that privacy risk is structurally eliminated rather than managed post hoc [79].

C. INFRASTRUCTURE, BUILDING, AND CITY-LEVEL TOOLS

From a GDPR perspective, infrastructure, building, and city-level carbon tools generally align with the principle of data minimization, as they primarily rely on aggregated energy consumption, material inventories, or urban-scale indicators rather than directly identifiable personal data [20]. However, when such tools integrate smart building systems, occupancy sensors, or location-based data, they may challenge purpose limitation and integrity/confidentiality

TABLE 10. Privacy evaluation of infrastructure, building, and city-level tools mapped against GDPR principles.

Tool	Privacy level	Reason
One Click LCA [71]	Low	Project-level data, no personal tracking
EC3 [71]	Low	Open material database, no PII collected
Tally [71]	Low	BIM plugin, local/project data only
ATHENA Impact Estimator [71]	Low	Building-level technical inputs only
City Energy Analyst [71]	Medium	Aggregate by default, but can ingest household data
Urban InVEST [71]	Low	Aggregated geospatial/land-use datasets
CITYgreen [71]	Low	GIS-based, public environmental data

principles, particularly if data collected for operational efficiency are later reused for benchmarking, policy enforcement, or third-party analytics without explicit safeguards [20], [21]. Infrastructure- and city-focused carbon tools generally pose a low privacy risk because they rely on technical, project, or geospatial data rather than personal identifiers. Tools such as One Click LCA, EC3, Tally, ATHENA, Urban InVEST, and CITYgreen operate with material databases, building models, or public spatial information, posing minimal risk of re-identification. The exception is City Energy Analyst, which, although often used with aggregated data, can also ingest household-level consumption or occupancy information, thereby increasing its risk level to medium. Overall, this category demonstrates that engineering and geospatial tools are less privacy-intrusive unless combined with fine-grained household datasets.

In addition to data aggregation and technical scope, the privacy impact of infrastructure, building, and city-level tools is shaped by the degree of institutional user agency embedded in their deployment. User agency in this context refers to the ability of municipalities, planners, or organizations to control data inputs, enforce aggregation thresholds, define retention periods, and prevent secondary use beyond the original planning or assessment purpose. Consistent with GDPR's principles of purpose limitation and privacy by design (Articles 5 and 25), tools that rely exclusively on public or project-level data afford strong governance control, whereas tools that permit ingestion of household- or occupancy-level datasets shift privacy responsibility to implementers and increase governance risk if safeguards are absent. Table 10 summarizes the privacy level of each tool in this category.

1) One Click LCA

One Click LCA operates at a low privacy risk by design, as it integrates with BIM software to analyze building materials and project specifications, including concrete volumes, insulation types, and structural components, without

collecting personal identifiers, household data, or behavioral patterns [71]. All inputs are technical, project-specific, and controlled entirely by the user or organization, with no automatic cloud syncing, third-party tracking, or commercial profiling [71]. Data remains in the owner's custody, stored locally or in secure, permissioned environments, and is used solely for lifecycle assessment and compliance with certification requirements. Its focus on architectural and construction-level metrics ensures that sustainability reporting occurs at the system level, not the human one. There is no inference of lifestyle, income, or identity; no surveillance of occupants or workers [71]. From a user agency perspective, One Click LCA affords strong control, as data entry is deliberate, project-scoped, and fully managed by the user or organization. There are no automated data feeds or opaque retention mechanisms, ensuring that users retain full authority over what is analyzed, stored, or shared [71].

2) EMBODIED CARBON IN CONSTRUCTION CALCULATOR

EC3 operates at a low privacy risk because it functions as an open, public database of embodied carbon for construction materials, allowing users to query emissions data by product type, manufacturer, or environmental product declaration without collecting any personal identifiers, user profiles, location data, or behavioral inputs [71]. Users interact with the tool by selecting materials from a transparent catalog; no login, tracking, or storage of individual queries is required by default. All data is sourced from verified Environmental Product Declarations, not from private user activity [71]. EC3's mission is transparency in material choice, not surveillance of users. Even when used within corporate workflows, it does not aggregate or monetize usage patterns. There is no profiling, no cloud-based identity linkage, and no secondary use of personal information [71]. Its power lies in democratizing access to environmental data, not in harvesting it. EC3 proves that profound sustainability impact can be achieved without compromising privacy: accountability begins with material truth, not personal tracking [71]. EC3 also demonstrates high user agency: users query a public database without contributing proprietary or identifiable data, and no persistent profiles or usage histories are imposed [71].

3) TALLY

Tally operates at a low privacy risk as a Revit plugin that performs lifecycle assessment directly within architectural design workflows, analyzing only building-specific technical data, such as material quantities, assembly types, and geometry, without accessing personal information, user identities, or behavioral logs [71]. All processing occurs locally on the designer's machine by default, with no automatic cloud uploads, account creation, or sharing of third-party data. Inputs are confined to project-level specifications, making it impossible to infer occupant habits, income, or demographics [71]. Tally's purpose is to embed sustainability into design, not to monitor its users. Even when exporting reports, data remains under the architect's control, with no

centralized storage or aggregation across projects. Its strength lies in seamless integration without intrusion: environmental accountability is achieved through precision in modeling, not surveillance in data collection [71]. User agency is particularly strong in Tally's local-first design, as architects maintain full control over project data, exports, and retention [71].

4) ATHENA Impact Estimator

ATHENA Impact Estimator operates at a low privacy risk by design, analyzing building-level engineering inputs, such as material volumes, insulation specs, and system efficiencies, to estimate lifecycle emissions, with no collection of personal, demographic, or behavioral data [71]. All inputs are technical, project-specific, and typically entered manually or imported from design files, remaining under the user's control without cloud storage, tracking, or third-party linkage. Its outputs serve planning and certification needs, not profiling, making re-identification of individuals or households impossible [71]. ATHENA focuses on systems, not people: it quantifies the environmental cost of construction, not the habits of occupants. With no user accounts, identifiers, or persistent data retention, it embodies ethical sustainability tooling, delivering rigorous analysis while safeguarding autonomy [71]. ATHENA further aligns with privacy-by-design by granting users complete autonomy over data entry and retention, with no embedded mechanisms for secondary use or cross-project aggregation [71].

5) CITY ENERGY ANALYST

The City Energy Analyst carries a medium privacy risk because, while its default use relies on aggregated, anonymized urban data, such as building stock typologies and district-level energy profiles, it can ingest high-granularity inputs, including household energy consumption, occupancy schedules, or socioeconomic indicators, if provided by users [71]. When such detailed datasets are imported, they enable precise modeling but also create opportunities for re-identification, behavioral profiling, or inference of sensitive traits (e.g., health conditions from heating patterns or income from appliance usage). Though City Energy Analyst itself doesn't collect data, its flexibility means privacy depends entirely on the user's input choices and local governance practices [71]. Without safeguards like anonymization, aggregation thresholds, or access controls, fine-grained imports transform it from a public planning tool into a potential surveillance vector. City Energy Analyst's power lies in its adaptability, but so does its vulnerability [71]. To remain ethically sound, its deployment must enforce strict data minimization: only use the granularity needed, never more. Transparency in input sourcing and clear policies on retention are essential to prevent well-intentioned analysis from becoming invasive monitoring [71]. From a user agency standpoint, City Energy Analyst represents a transitional case: while the tool itself does not mandate intrusive data collection, it places responsibility on users to enforce aggregation, anonymization, and access controls when importing

fine-grained datasets [71]. In the absence of institutional governance, this flexibility can unintentionally undermine privacy safeguards.

6) Urban InEST

Urban InVEST operates at a low privacy risk by relying exclusively on aggregated, public geospatial datasets, such as land cover, canopy density, soil type, and census-level demographics, without accessing or processing any personal identifiers, individual behaviors, or private property data [71]. It models ecosystem services and carbon sequestration at neighborhood or municipal scales, using satellite imagery and open administrative boundaries that cannot be traced to individuals [71]. Its design prioritizes environmental insight over individual surveillance: it answers, “how much green space stores carbon?” not “who lives where and how they live.” No user accounts, location tracking, or behavioral logging occur, and all outputs are statistical, never attributable to specific households or residents [71]. Even when combined with socioeconomic indicators, data is anonymized and aggregated to prevent re-identification. Urban InVEST exemplifies ethical urban modeling, empowering planners to enhance climate resilience without compromising civil privacy. Its strength lies in scale without intrusion: sustainability achieved through landscape wisdom, not personal surveillance [71]. Urban InVEST affords strong user agency by restricting analysis to public and aggregated datasets, eliminating opportunities for scope creep or unauthorized secondary use [71].

7) CITYgreen

CITYgreen operates at a low privacy risk by analyzing only public, environmental data, such as tree species, canopy cover, and land use, through GIS mapping, with no collection of personal identifiers, residential addresses, or behavioral patterns [71]. It quantifies carbon storage and cooling benefits at the neighborhood or city scale, using open datasets such as municipal tree inventories and satellite imagery, ensuring that outputs are aggregated and non-attributable. Its focus is on infrastructure, not individuals: it measures trees, not people. No user accounts, location tracking, or data uploads are required, and all inputs derive from publicly accessible or municipally shared sources [71]. Even when used to target equity gaps in green space, it relies on census tracts of household-level details, preserving anonymity. CITYgreen proves that meaningful urban climate action can be driven by ecological insight alone, without compromising privacy [71]. CITYgreen similarly exhibits high user agency, as it relies exclusively on public GIS layers and municipal datasets, with no mechanisms for ingesting private or household-level information [71].

D. IoT, AI, BLOCKCHAIN, AND ADVANCED FRAMEWORKS

IoT- and blockchain-based tools present particular challenges for data minimization and the right to erasure, as real-time data streams and immutable ledgers conflict with GDPR requirements for controllability and storage limitation

TABLE 11. Privacy evaluation of IoT, AI, blockchain, and advanced frameworks mapped against GDPR principles.

Tool	Privacy level	Reason
IoT Sensors / Smart Meters [70, 75]	High	Passive real-time sensing, highly granular.
Blockchain Carbon Tracking [22, 80]	High	Immutable transparency risks without privacy tech.
Digital Product Passport [77]	Medium	Product level sensitive but not personal.
UG-CET Multi-Chain [26]	High	Global transaction data, commercially sensitive.
zk-SNARKs [55]	Low	Verifies claims without revealing raw data.
DP-FedSW [82]	Low	Local training with DP; raw data not shared.
Smart Grid AI Systems [75]	High	High-frequency household → de-anonymization risk.

[20]. While such tools enhance transparency, they may violate privacy-by-design principles unless combined with privacy-preserving mechanisms such as zero-knowledge proofs or off-chain processing. Tools in this category show the sharpest divide between high-risk and privacy-preserving approaches. IoT sensors, Smart Meters, Blockchain Carbon Tracking Platforms, UG-CET, and Smart Grid AI systems pose a high privacy risk, as they rely on highly granular, real-time, or immutable data that can expose household routines, financial transactions, or sensitive operational details. The Digital Product Passport is rated medium risk, since it focuses on product-level information that can still reveal commercially sensitive data. By contrast, zk-SNARK protocols and DP-FedSW demonstrate how cryptographic and federated approaches can enable carbon reporting with minimal disclosure, keeping them at low risk.

In this category, user agency is often structurally constrained by design, rather than merely limited by implementation. Continuous data streams, automated ingestion, and immutable ledgers significantly restrict the ability of individuals or organizations to exercise meaningful consent, withdraw participation, or enforce deletion once data collection begins. This directly challenges GDPR principles related to controllability, storage limitation, and privacy by design (Articles 5, 7, 17, and 25) [20]. As a result, privacy risk in IoT-, AI-, and blockchain-based carbon tools is driven not only by data sensitivity and granularity, but by the extent to which users retain, or permanently lose, governance over their data lifecycle. Table 11 summarizes the privacy level of each tool in this category.

1) IoT SENSORS AND SMART METERS

IoT Sensors and Smart Meters pose a high privacy risk due to their passive, continuous collection of granular, real-time

data, such as minute-by-minute electricity use, CO₂ levels, occupancy patterns, and device-level GPS, that can infer intimate details: when residents sleep, cook, or use medical equipment; whether a business operates nights; or even health conditions from appliance cycles [70]. This data is routinely transmitted to centralized cloud or utility servers, where it's vulnerable to breaches, insider access, or undisclosed secondary uses, including insurance underwriting, targeted advertising, or law enforcement surveillance [75]. Even anonymized streams can be re-identified through unique behavioral signatures. These tools turn homes and workplaces into surveillance environments under the guise of efficiency [70]. While they enable energy savings, they also normalize pervasive monitoring without meaningful consent or control. The risk isn't theoretical, it's operational [75]. From a user agency perspective, these systems offer minimal control once deployed: consent is often bundled into utility contracts, data collection is continuous, and users lack practical mechanisms to pause monitoring, restrict granularity, or delete historical records [75]. This absence of meaningful opt-out or data sovereignty substantially amplifies privacy risk beyond technical exposure alone.

2) BLOCKCHAIN CARBON TRACKING SYSTEMS

Blockchain Carbon Tracking poses a significant privacy risk due to its core design: emissions data, including supply chain flows, production volumes, and energy sources, is permanently recorded on an immutable ledger, thereby exposing sensitive operational details to all participants [80]. Even on permissioned chains, the transparency that ensures accountability also enables competitors, regulators, or bad actors to infer proprietary strategies, supplier relationships, or efficiency benchmarks through pattern analysis [22]. Once on-chain, data cannot be deleted or anonymized effectively without cryptographic safeguards like ZKPs or off-chain storage with on-chain hashes. The permanence of blockchain turns confidentiality into a liability: every verified emission claim becomes a permanent public record [80]. Secondary use risks include corporate espionage, reputational targeting, or regulatory overreach. While ZKPs and hybrid architectures can mitigate exposure, most implementations lack them by default [22]. User agency is fundamentally constrained by blockchain immutability: once emissions data are recorded, organizations cannot retract, modify, or selectively erase records, even if contextual risks change over time [22]. This irreversibility conflicts directly with GDPR's right to erasure and future-proof privacy expectations, intensifying governance risk unless cryptographic safeguards are embedded by design.

3) DIGITAL PRODUCT PASSPORT

The Digital Product Passport carries a moderate privacy risk, as it stores supply-chain carbon metadata, including material origins, manufacturing processes, and emissions intensity, on shared, often blockchain-based ledgers [77]. While no personal data is collected, this information can

reveal commercially sensitive operational details: proprietary production techniques, supplier networks, cost structures, or competitive benchmarks that, if exposed, could undermine market advantage [77]. Though permissioned access and selective disclosure mechanisms help limit visibility to authorized stakeholders, the permanence of ledger entries means any disclosed data remains permanently traceable and potentially constructible through aggregation or cross-referencing. Digital Product Passports enhance transparency for compliance and certification, but their design inherently trades confidentiality for trust. Without robust encryption, zero-knowledge proofs, or off-chain storage for sensitive fields, even well-intentioned implementations risk exposing corporate sustainability reporting [77]. The medium risk reflects this balance: not invasive of individuals, but vulnerable to industrial espionage. To uphold both accountability and competitiveness, Digital Product Passport systems must embed privacy-by-design, not as an add-on, but as a core requirement [77]. Although access controls can limit who views passport data, user agency remains partial, as data contributors cannot fully revoke or expire disclosed information once it is written to shared ledgers [77]. This limited reversibility justifies its medium-risk classification when governance and long-term control are considered.

4) UG-CET

UG-CET carries a high privacy risk due to its global, multi-blockchain architecture that records transaction-level details, such as emission allowance volumes, trader identities, facility locations, and pricing, across jurisdictions on immutable ledgers [26]. This granular, permanent record exposes not only financial flows but also proprietary operational data: production capacities, compliance strategies, and supply chain dependencies that competitors or regulators could exploit. Even pseudonymous entities can be re-identified through pattern analysis, cross-referencing with public filings, or linking to satellite or trade data [26]. The system's strength, global transparency, is also its greatest vulnerability: every trade becomes a public footprint, eroding corporate confidentiality and inviting market manipulation or geopolitical targeting. Secondary use risks include commercial profiling, regulatory overreach, or unintended exposure of developing economies' emissions profiles. While decentralization enhances integrity, it eliminates centralized control over data access or deletion [26]. Organizational user agency is particularly weak in UG-CET, as decentralized consensus mechanisms eliminate any central authority capable of enforcing selective disclosure, deletion, or contextual access restrictions. Once transaction data enter the system, participants permanently relinquish control, elevating privacy risk from both a technical and governance perspective [26].

5) ZERO-KNOWLEDGE PROOF PROTOCOL

zk-SNARKs operate at a low privacy risk by design, enabling organizations to cryptographically prove the accuracy of emissions data, such as compliance with targets or offset

validity, without revealing any underlying raw inputs, including energy use, supplier details, or production volumes [55]. Built on mathematical guarantees, zk-SNARKs verify truth while preserving secrecy, eliminating exposure risks inherent in traditional transparent ledgers. This is not mere obfuscation; it's principled confidentiality: corporate sensitive data remains entirely private, yet auditors, regulators, and markets can still trust the outcome [55]. Unlike standard blockchain systems that broadcast data, zk-SNARKs ensure no metadata leaks, no patterns are inferable, and no re-identification is possible [55]. Importantly, zk-SNARK-based systems maximize user agency by allowing organizations to choose what is proven without ever disclosing what is known [55]. Data never leaves the owner's control, and no persistent records of sensitive inputs are created, fully aligning with privacy-by-design and user sovereignty principles.

6) DP-FedSW FRAMEWORK

DP-FedSW operates at a low privacy risk by design, as it enables collaborative carbon modeling across industrial sites without centralizing raw data. Each participant trains models locally, and only noisy, privatized model updates (not individual measurements) are shared [82]. Differential privacy injects calibrated noise into these updates, mathematically guaranteeing that no individual facility's operational details, such as energy use patterns or production volumes, can be inferred or re-identified from aggregated outputs [82]. This architecture transforms collective insight into a privacy-preserving achievement, enabling organizations to jointly optimize emissions reductions while safeguarding proprietary information, maintaining competitive advantage, and ensuring regulatory compliance [82]. Unlike cloud-based analytics that demand data surrender, DP-FedSW ensures data sovereignty remains with the source. There is no central database to breach, no user profiling possible, and no secondary misuse of granular inputs. It exemplifies ethical scalability, where sustainability collaboration thrives not through exposure but through encrypted cooperation [82]. DP-FedSW similarly embeds strong user agency by design: participants retain full ownership of raw data, control model participation, and can exit collaborative processes without exposing historical records [82]. This reversibility and local control distinguish it sharply from centralized AI analytics.

7) SMART GRID AI SYSTEMS

Smart Grid AI Systems carry a high privacy risk due to their reliance on high-frequency, granular telemetry from smart meters and IoT devices that capture minute-by-minute energy usage patterns across homes and businesses [75]. This data reveals intimate behavioral signatures: when residents wake, sleep, cook, use medical devices, or are away; whether a factory runs night shifts; or if a home uses heat pumps versus gas heating, all of which can infer health conditions, income levels, routines, and even political or religious habits [75]. Even when anonymized, such fine-grained time-series data is highly susceptible to re-identification through pattern

matching with public datasets or cross-referencing with other sources [75]. The very efficient gains from predictive optimization come at the cost of pervasive surveillance: these systems turn the electrical grid into a behavioral monitoring network. Centralized storage of this data creates attractive targets for breaches, and secondary uses, including insurance underwriting, targeted advertising, or law enforcement access, are not only plausible but economically incentivized [75]. User agency is weakest in smart grid AI deployments, where participation is often mandatory, data collection is continuous, and withdrawal is impractical without disconnecting from essential services [75]. The lack of granular consent, data portability, or deletion options reinforces their classification as high-risk under the user agency criterion.

E. EXPERIMENTAL AND RESEARCH-ORIENTED TOOLS

Most experimental and research-oriented carbon footprint tools have low privacy risk, as they primarily process technical or system-level data. Tools like Carbon Tracker, Green Algorithms, Experiment Impact Tracker, ML CO2 Impact, and Cumulator rely on local monitoring or manual technical inputs without collecting personal identifiers. Digital Services Carbon Models also remain at low risk, as they use aggregated service benchmarks rather than individual logs. Higher risks arise with the CO2web Observatory and COMPAIR Dashboard, which accept personal or household-level inputs for calculators and simulations. If such data are retained or shared, these tools are classified as having a medium privacy risk.

From a user agency perspective, experimental and research-oriented tools generally perform well, as participation is voluntary, data inputs are user-controlled, and most tools operate locally without persistent storage. Users typically retain full discretion over what data is entered, whether outputs are shared, and when tools are used or discontinued. Consequently, privacy risk in this category is driven less by automated surveillance and more by governance choices around optional data retention, research reuse, and transparency of experimental protocols. Table 12 summarizes the privacy level of each tool in this category.

1) CARBON TRACKER

Carbon Tracker maintains a low privacy impact by design, avoiding all personal or identifiable data and processing only raw hardware metrics locally, without relying on cloud services, user accounts, or third-party integrations [49]. No behavioral tracking, geolocation, or network activity is monitored; only power consumption and runtime are measured, with results stored entirely on-device. This eliminates risks of profiling, re-identification, or unauthorized access, even if data is shared for reproducibility [49]. Unlike tools that infer habits from usage patterns, it remains agnostic to who is running the code, focusing solely on what the machine consumes. Its ethical strength lies in this restraint: accountability is achieved through technical precision, not surveillance. By refusing to collect more than necessary,

TABLE 12. Experimental and research-oriented tools for privacy impact.

Tool	Privacy level	Reason
Carbon Tracker [49]	Low	Local telemetry only, no identifiers.
Green Algorithms [49]	Low	Manual technical inputs, no PII.
Experiment Impact Tracker [49]	Low	Local monitoring, system-level only.
ML CO ₂ Impact [49]	Low	Manual online inputs, no persistence.
Cumulator [49]	Low	Local estimation, system data only.
CO ₂ web Observatory [50]	Medium	Neutral product DB, personal inputs may be stored.
COMPAIR Dashboard [41]	Medium	Household-level scenario inputs storage risks.
Digital Services Carbon Models [69]	Low	Aggregate modeling, no PII.

it models responsible innovation, where sustainability doesn't demand surrender of privacy. Carbon Tracker proves that impactful climate action in computing can, and should, coexist with digital autonomy [49]. User agency is maximized, as all data collection occurs locally, is fully transparent, and can be stopped or deleted at any time by the user [49]. No irreversible records are created, and participation leaves no residual digital footprint beyond what the user chooses to share.

2) GREEN ALGORITHMS

Green Algorithms operates at a low privacy risk as a web-based calculator requiring only manual, technical inputs, such as runtime, CPU/GPU cores, and cloud region, with no collection of personal identifiers, location data, device fingerprints, or behavioral tracking [49]. All calculations are performed on the client side or using publicly available emission factors, with no user accounts, data storage, or server-side profiling. Users retain complete control over what they enter, and no information is retained after submission [49]. Its purpose is educational transparency, not surveillance: it empowers researchers to understand computational emissions without surrendering privacy. There is no mechanism for re-identification, no third-party sharing, and no secondary use of data [49]. From a user agency standpoint, Green Algorithms offers complete control: inputs are optional, ephemeral, and never retained. Users can freely explore scenarios without any obligation to disclose identifiable or persistent information [49].

3) EXPERIMENT IMPACT TRACKER

Experiment Impact Tracker operates at a low privacy risk by design: it runs locally on the user's machine, collecting only technical telemetry, such as CPU/GPU power draw,

memory usage, and runtime, during computational experiments [49]. It accesses no personal files, user identities, IP addresses, network activity, or external data sources. All measurements are confined to system-level metrics required for energy estimation, with no cloud upload, account creation, or third-party transmission [49]. Its architecture prioritizes privacy-first computing: users retain complete control over their data, which is never stored beyond local use unless explicitly exported. There is zero profiling, no behavioral tracking, and no risk of re-identification [49]. User agency remains high, as researchers can start, stop, export, or discard measurements at will, with no external dependencies or automatic data transmission [49].

4) ML CO₂ IMPACT

ML CO₂ Impact operates at a low privacy risk by design, as it requires only manual and technical inputs, such as hardware type, runtime duration, and geographic region, with no tracking, user accounts, cookies, or persistent storage [49]. All calculations occur client-side using public grid intensity data, and no personal identifiers, location history, device fingerprints, or behavioral patterns are collected. Users retain full control over what they enter, with zero telemetry, cloud uploads, or secondary data use [49]. Its purpose is transparency without intrusion: empowering researchers to estimate computational emissions through voluntary, anonymous input. There is no profiling, re-identification is not possible, and no mechanism for surveillance exists [49]. This tool exemplifies consent-based engagement: users decide what to enter, no background data is collected, and no records persist beyond the session. This aligns strongly with GDPR principles of controllability and storage limitation [49].

5) CUMULATOR

Cumulator operates at a low privacy risk as a local Python package that estimates energy consumption from computational and network activity, such as data transfer volumes and GPU/CPU usage, using only system-level telemetry [49]. It requires no user registration, does not access personal files or identifiers, and transmits no data externally. All processing occurs offline on the user's device, ensuring complete control over input and output [49]. Its design prioritizes ethical research: it quantifies the environmental cost of distributed ML without compromising confidentiality or introducing surveillance. There is no profiling, no behavioral tracking, and no risk of re-identification [49]. Cumulator preserves strong user agency by remaining fully offline and user-operated. Researchers maintain exclusive control over inputs, outputs, and any downstream use of results, with no embedded data extraction mechanisms [49].

6) CO₂web Observatory

CO₂web Observatory (research component) carries a medium privacy risk due to its dual functionality: while its product emissions database is neutral and public, the personal calculator allows users to input detailed lifestyle

data, such as travel, diet, or spending, which, if stored persistently or linked to profiles, can form sensitive behavioral patterns revealing income, habits, or ethics [50]. Although no automatic tracking occurs, the potential for re-identification increases if data is retained server-side, aggregated across users, or shared for research without explicit anonymization. The tool's value lies in connecting individual choices to systemic impact, but this very feature creates ethical tension: awareness must not come at the cost of silent data harvesting [50]. Privacy risk is contingent on implementation: with local-only storage and opt-in consent, it remains low; with centralized retention or undisclosed secondary use, it escalates. To uphold trust, researchers using CO2web must enforce strict data minimization, clear retention policies, and user-controlled deletion, ensuring transparency serves sustainability, not surveillance [50]. User agency in CO2web depends heavily on implementation choices: while data entry is voluntary, agency is reduced if users lack clear visibility into retention periods, reuse for research, or deletion mechanisms. Transparent consent flows and opt-out options are therefore essential to prevent educational tools from undermining participant autonomy [50].

7) COMPAIR DASHBOARD

The COMPAIR Dashboard poses a medium privacy risk because, although designed for participatory policymaking, it often requires users to input household or location-specific data, such as energy use, vehicle ownership, or income brackets, to personalize scenario simulations [41]. Though not inherently invasive, these inputs can reveal sensitive socioeconomic or behavioral patterns when stored centrally or aggregated across users. Without differential privacy, anonymization, or strict access controls, such data may be re-identified or exploited for profiling, marketing, or unintended research purposes [41]. The tool's strength in fostering civic engagement is matched by its potential for subtle surveillance: the more personalized the simulation, the greater the risk of turning public participation into private data extraction. The risk remains medium because inputs are voluntary and typically non-financial; however, without transparent policies on retention, sharing, and deletion, trust erodes [41]. Although participation is voluntary, user agency may be weakened if contributors are unaware of how long their inputs are stored or whether they are reused beyond the immediate simulation [41].

8) DIGITAL SERVICES CARBON MODELS

Digital Services Carbon Models operate at a low privacy risk by relying on top-down, aggregate metrics, such as average data transfer volumes, server utilization rates, or regional grid intensities, to estimate emissions from digital services like streaming or cloud computing [69]. These models use benchmarked, population-level assumptions rather than individual user logs, device identifiers, or behavioral data, ensuring no PII is ever collected, stored, or processed [69]. Their strength lies in scalability and generalizability: they quantify systemic

impact without intruding into personal behavior. There is no tracking, no profiling, and no storage of user activity, only public, averaged technical parameters. This makes them ideal for research, policy, and industry standards where environmental accountability must coexist with digital rights [69]. User agency concerns are minimal in this category, as no individual participation or data contribution is required [69].

F. COMPARISON OF PRIVACY RISKS ACROSS CATEGORIES

The privacy risk comparisons herein are derived from an analysis of published data practices and system architectures, not from hands-on empirical testing or examination of proprietary implementations. Conducting a cross-analysis of carbon footprint tools across categories is valuable because it highlights both the best practices and the most significant risks in managing privacy [57]. While individual tools vary in design and purpose, comparing them side by side allows us to identify common patterns that influence privacy outcomes. By highlighting the least risky tools, we can demonstrate approaches that strike a balance between accuracy and confidentiality. At the same time, highlighting the riskiest tools draws attention to where safeguards are urgently needed [57]. This dual perspective strengthens recommendations for future tool development and responsible adoption.

1) CROSS-CATEGORY COMPARISON OF PRIVACY RISKS

The evaluation of carbon footprint tools across the five categories shows a spectrum of privacy risks that directly correlate with the type of data collected, its granularity, and the extent of external processing or sharing. These risks are further shaped by the degree of user agency embedded in each tool, particularly whether data provision is voluntary, granular consent is available, and users retain control over storage, access, and deletion. Figure 2 summarizes the general findings among the privacy levels of tools across the five categories. At one end of the spectrum, individual and household calculators are the most privacy-sensitive category. Simple web-based calculators, such as WWF and MyClimate, remain at low risk. Still, once tools integrate real-time sensing, GPS tracking, or bank transaction data, as in EcoTrack, ECOSAUR, Tanaffas, or Doconomy, the privacy risks escalate to an unacceptable level. These tools often combine continuous data collection with sensitive identifiers, making them intrusive and vulnerable to profiling or misuse. A key differentiator within this category is user control: tools relying on passive sensing or financial integration sharply reduce user agency compared to calculators based on voluntary, one-time inputs.

By contrast, corporate and organizational tools generally fall into the low to medium range. Research-focused utilities like GES 1point5, CodeCarbon, EmissionsTracker, Green Metrics Tool, and FootPrinter pose minimal risks because they process aggregated or technical data locally. However, when organizations adopt third-party platforms such as Carbon Footprint Ltd., Carbon Trust tools, or Greenhouse Gas Protocol implementations, they expose operational details

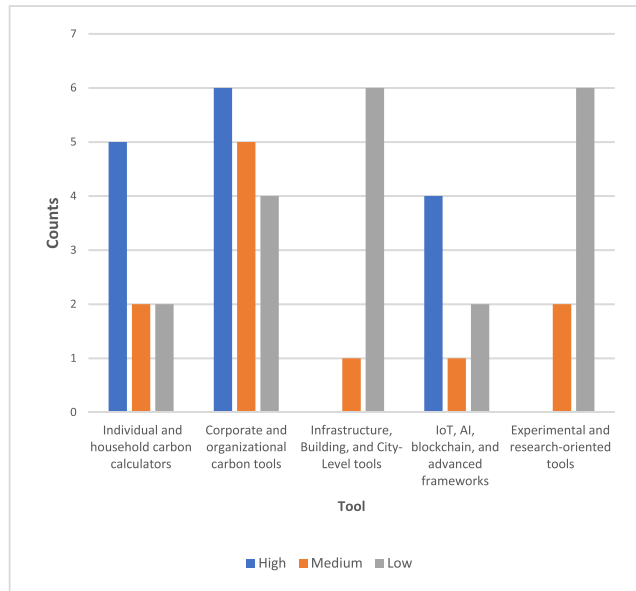


FIGURE 2. Number of high, medium, and low privacy impact tools in each category.

to external actors, thereby raising medium-level risks. Tools like Carbon Analytics Dashboards, Microsoft Cloud for Sustainability, Salesforce Net Zero Cloud, SAP Sustainability Control Tower, and IBM Environmental Intelligence Suite are exceptions, classified as high risk, since they directly integrate with sensitive financial and operational systems.

Infrastructure, building, and city tools demonstrate the lowest average privacy risks, as they rely heavily on engineering and geospatial datasets rather than personal identifiers. Tools like One Click LCA, EC3, Tally, ATHENA, Urban InVEST, and CITYgreen are consistently low risk. The only exception is City Energy Analyst, which can ingest household-level consumption or occupancy data, raising its classification to medium risk. This category shows how privacy risks emerge only when aggregated environmental data are combined with fine-grained personal or household datasets.

The IoT, AI, and blockchain tools represent a sharp divide. On one side, IoT sensors, smart meters, blockchain platforms, UG-CET, and smart grid AI systems pose high privacy risks due to their reliance on real-time telemetry, immutable transaction trails, and highly granular operational data. In these systems, privacy risk is amplified not only by data granularity but also by the absence of meaningful user control, as continuous collection and immutability eliminate practical options for consent withdrawal or data erasure. On the other hand, advanced privacy-preserving frameworks, such as zk-SNARKs and DP-FedSW, demonstrate that it is possible to enable carbon accounting while safeguarding confidentiality. These fall into the low-risk category, highlighting a promising direction for future tool development.

Finally, experimental and research-oriented tools mostly fall into the low privacy risk category. Tools like Carbon Tracker, Green Algorithms, and ML CO2 Impact collect only

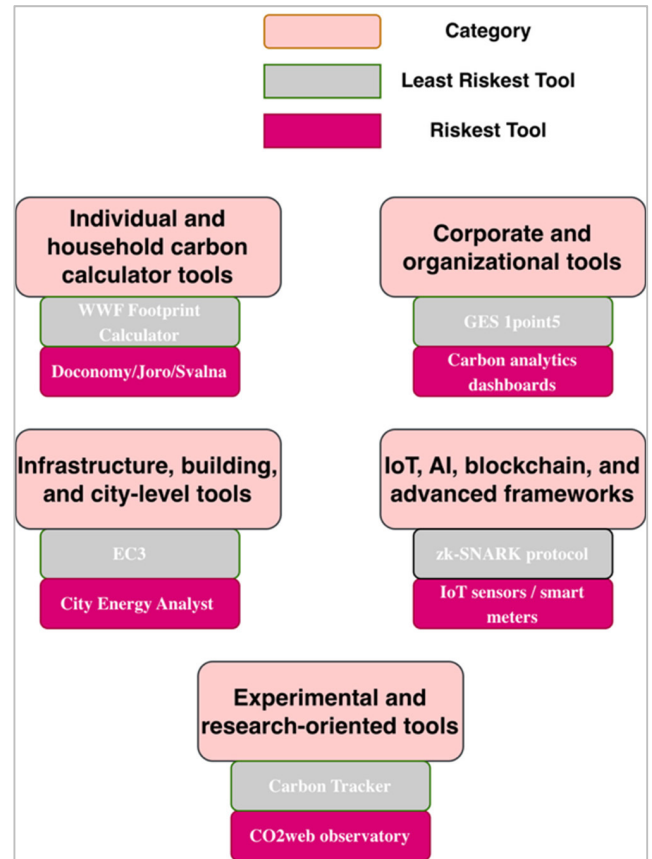


FIGURE 3. Privacy is the highest and least risky tool in each category.

system-level or manually entered technical data. Their design prioritizes academic reproducibility and awareness rather than large-scale data collection. The CO2web Observatory and COMPAIR Dashboard are the exceptions, both rated as medium risk due to their reliance on user-supplied personal or household data, which could be retained or shared.

2) PRIVACY LEAST VERSUS RISKY TOOLS

When comparing carbon footprint tools across categories, a key finding is that not all solutions carry the same privacy burden. Each category contains at least one tool that demonstrates how carbon accounting can be achieved while minimizing risks to user confidentiality. Figure 3 summarizes the riskiest and least risky tools across the five categories.

In the Individual & Household Calculators category, the WWF Footprint Calculator stands out as the least risky [85]. It collects only voluntary, self-reported lifestyle information through a simple questionnaire, without storing sensitive identifiers or relying on third-party integrations [68]. This minimalistic approach ensures that privacy risk remains low while still raising awareness of personal emissions. Within Corporate & Organizational Tools, GES Ipoint5 demonstrates strong privacy-by-design [45]. It focuses on aggregated and anonymized organizational reporting, adhering to GDPR standards to ensure that data remains non-identifiable [45]. Unlike commercial dashboards or consultancy-hosted

platforms, it avoids unnecessary exposure of sensitive operational information, making it the least risky option in this category.

For Infrastructure, Building & City Tools, several tools are equally low risk, but is exemplary. EC3 utilizes an open-access database of construction materials and embodied carbon factors, eliminating the need for personal or household data [71]. This makes it a strong model of a low-risk, technical tool. In the IoT, AI, and Blockchain category, most solutions are high risk, but zk-SNARKs provide a notable counterexample. These cryptographic protocols enable organizations to verify the accuracy of carbon accounting claims without disclosing raw data [55]. By design, they prioritize confidentiality, making zk-SNARKs the least risky in this otherwise high-risk category. Finally, in the Experimental & Research-Oriented Tools category, the Carbon Tracker package is the least risky. It measures power consumption and runtime locally on users' machines, never collecting personal identifiers or transmitting data externally [49]. This makes it particularly suitable for academic research environments, where privacy and reproducibility must both be preserved.

It is also beneficial to identify the riskiest tools from all five categories. Our assessment remains based on the five key criteria: the type of data collected, its granularity, how it is stored and processed, its potential for secondary use or misuse, and user agency. Across categories, tools identified as least risky consistently preserve higher user agency, whereas the riskiest tools minimize or eliminate user choice over data collection, retention, and reuse. In the Individual & Household Calculators category, the riskiest tools are those that integrate with financial institutions. Doconomy (and similar apps like Joro and Svalna) directly link to users' bank or credit card transactions to calculate their footprints [53]. This exposes highly sensitive, transaction-level data, including purchasing habits, locations, and potentially health-related activities [52]. Such granularity and centralization make Doconomy the highest risk in this category.

For corporate & organizational tools, the riskiest solution is Carbon Analytics Dashboards. These tools integrate directly with corporate finance and operations systems to automate carbon reporting [41]. By doing so, they handle sensitive information, including supplier payments, employee travel logs, and production costs. If breached, this data could expose trade secrets and competitive intelligence, placing Carbon Analytics at a high privacy risk level. In infrastructure, building & city tools, most are low risk, but the City Energy Analyst can become problematic. While often used with aggregated datasets, it is capable of ingesting detailed household-level consumption and occupancy data for modeling [71]. If such inputs are stored or shared, they can expose sensitive behavioral patterns, making City Energy Analyst the highest risk in this otherwise low-risk category.

Among IoT, AI, and Blockchain tools, the riskiest are IoT sensors and smart meters [75]. Their continuous, high-frequency monitoring of energy use, appliance-level data, or GPS signals generates detailed profiles of household

routines, occupancy, and even personal habits [70]. Since the data are usually stored centrally, they carry enormous risks of surveillance and misuse, making IoT sensors the highest risk. Finally, in the experimental & research-oriented tools category, most solutions are low risk, but the CO2web Observatory stands out as more intrusive [50]. Although it primarily hosts a product emissions database, its personal carbon calculator requires users to enter lifestyle and consumption data [50]. If such data is retained or shared, users can be profiled. While less risky than IoT or finance-linked tools, within this category CO2web Observatory is the riskiest.

3) CROSS-CATEGORY SYNTHESIS AND TRADE-OFFS

Viewed collectively, the five categories reveal consistent cross-cutting patterns that transcend individual tool designs. First, privacy risk increases systematically with data automation and temporal resolution: tools based on manual input or aggregated technical data (e.g., WWF Calculator, EC3, CodeCarbon) remain low risk, whereas real-time sensing, financial integration, and immutable ledgers (e.g., smart meters, finance-linked apps, blockchain platforms) introduce persistent exposure and re-identification risks [85]. Second, the primary privacy trade-off is not between sustainability intent and misuse, but between accuracy and controllability: higher precision typically requires continuous, fine-grained data that reduces user or organizational agency [44], [45]. Third, risk is strongly context-dependent: household-facing tools disproportionately affect personal autonomy, while enterprise platforms concentrate risk around confidentiality, competitive intelligence, and data sovereignty. Finally, the comparison highlights a clear design pathway forward. That is, privacy-preserving architectures such as federated learning, differential privacy, and zero-knowledge proofs consistently enable cross-category carbon accounting without proportional increases in privacy risk [70], [75]. These patterns suggest that privacy risk in carbon footprinting is driven less by sector and more by architectural choice.

G. SECTORAL CONTEXT AND CONTEXTUAL APPROPRIATENESS OF PRIVACY RISKS

While this study applies a uniform privacy evaluation framework across all categories of carbon footprint tools, it is important to acknowledge that privacy expectations and acceptable risk thresholds vary significantly by sector and use context. Privacy is not an absolute concept; rather, it is shaped by social norms, regulatory environments, and the balance between public interest and individual or organizational rights [99]. As such, a tool classified as "high privacy risk" under the proposed framework may be perceived and governed differently depending on its sectoral application.

In household and individual carbon tracking, privacy expectations are typically high, as tools often collect data related to personal lifestyles, mobility patterns, energy consumption, or financial transactions [85]. In these contexts, granular monitoring and centralized storage may be viewed

as intrusive, and even moderate data misuse could undermine user trust [67]. Consequently, tools operating in this domain require particularly strong safeguards aligned with consent, data minimization, and purpose limitation principles.

By contrast, industrial and large-scale emissions reporting often operates under a different normative and regulatory logic [44]. Emissions data in energy, manufacturing, and transportation sectors are frequently subject to public disclosure requirements, regulatory audits, and ESG transparency obligations [45]. In such settings, more intrusive data collection may be considered socially acceptable or even necessary, provided it is accompanied by institutional safeguards, accountability mechanisms, and clear legal mandates.

Small and medium-sized enterprises (SMEs) represent an intermediate case. While SMEs may not face the same public disclosure pressures as large corporations, they often possess limited resources to manage complex data governance frameworks [71]. Here, privacy concerns are closely linked to competitive sensitivity, as detailed emissions or operational data could reveal trade secrets or market positioning.

Accordingly, this study's privacy risk classifications should be interpreted as baseline technical and governance assessments, rather than absolute judgments. A key implication is the need for contextual appropriateness, whereby privacy safeguards, governance models, and acceptable trade-offs are calibrated to sector-specific norms, regulatory obligations, and stakeholder expectations. Future frameworks may therefore benefit from integrating sector-aware weighting mechanisms that reflect the varying balance between transparency, accountability, and privacy across different application domains.

VI. DISCUSSIONS

The present study systematically reviewed the literature to investigate a central research question: RQ, evaluate, and compare the privacy risks of carbon footprint measurement.

A. PRACTICAL IMPLICATIONS

The cross-category comparison reveals a key insight: privacy risk increases with the granularity and automation of data collection. An equally important factor is the degree of user agency embedded in these tools, particularly whether individuals or organizations retain meaningful control over when data is collected, how it is processed, and whether it can be withdrawn or deleted. Manual-entry calculators and technical research tools remain at low risk, while tools that embed in real-time sensing, financial transactions, or immutable ledgers pose the highest risks. Categories such as infrastructure/building tools demonstrate that effective carbon accounting can be achieved with minimal privacy intrusion. In contrast, IoT and household calculators highlight the urgent need for privacy-preserving safeguards. These findings underscore that the future of carbon footprint tools depends on reconciling data accuracy with user confidentiality through mechanisms such as anonymization, cryptographic proofs, and federated learning.

A core tension across carbon tools is the trade-off between accuracy and privacy: high-accuracy tools (e.g., IoT sensors, finance apps) reveal intimate behavioral patterns at significant privacy cost, while low-risk tools (e.g., WWF Calculator, CodeCarbon) sacrifice precision for autonomy. The former enables dynamic insights but risk surveillance; the latter protects dignity but lacks granularity. This trade-off is evident when comparing individual household tools. The WWF Footprint Calculator preserves privacy by relying on coarse, self-reported inputs that are neither stored nor linked to external systems, resulting in low privacy risk but approximate emissions estimates. In contrast, finance-linked tools such as Doconomy achieve higher accuracy by integrating transaction-level financial data, but at the cost of exposing highly granular behavioral and financial patterns through centralized processing. This trade-off is not only technical but also procedural: high-risk tools often minimize user choice through passive sensing or default data capture, whereas low-risk tools preserve agency through voluntary input, transparency, and user-controlled data lifecycles. Bridging this gap requires ethical innovation; privacy-preserving techniques like differential privacy, federated learning, and zero-knowledge proofs can deliver accurate, scalable emissions tracking without exposing personal data. Sustainability must not demand surrender of privacy. The future lies not in choosing between accuracy and anonymity, but in designing systems that achieve both through cryptographic and architectural integrity.

A major challenge in carbon footprinting is fragmentation: corporate tools such as the Greenhouse Gas Protocol, One Click LCA, and EC3 align with global standards, ensuring credible and comparable ESG reporting. In contrast, consumer-facing apps often employ opaque, custom methodologies with inconsistent emission factors, resulting in wildly varying results for the same activity. This lack of harmonization undermines trust, confuses users, and hinders the ability to conduct meaningful benchmarking. Without standardized, open datasets and transparent calculation frameworks, comparability remains elusive, especially across sectors and geographies. Achieving true accountability requires global alignment: open-access methodologies, shared databases, and interoperable protocols must be adopted universally to ensure that every footprint, whether measured by a corporation or a citizen, speaks the same language as environmental truth.

A growing cohort of tools, like Carbon Tracker, CodeCarbon, and EmissionsTracker, is addressing the rising carbon footprint of digital services, AI training, and cloud computing. As digitalization accelerates, quantifying the energy use of algorithms and infrastructure has become essential for sustainable tech. Unlike consumer-facing apps, these tools rely on technical metrics, hardware usage, runtime, power draw, with no personal data collected, ensuring low privacy risk. Their localized, open-source nature further enhances trust and ethical adoption. This makes them uniquely positioned to scale climate accountability in tech without compromising

autonomy. By making invisible emissions visible at the system level, they offer a rare win-win: precise environmental insight with minimal ethical cost, paving the way for responsible innovation in an increasingly digital world.

The shift from manual calculators to automated, real-time monitoring marks a transformative leap in carbon tracking. Modern tools now leverage IoT sensors, smart meters, and AI analytics to deliver dynamic, predictive insights, optimizing energy use in homes, factories, and grids. Blockchain further enables transparent supply chain tracing and carbon credit verification. While these advances boost accuracy and responsiveness, they also intensify privacy and cybersecurity risks: continuous data streams reveal intimate behavioral patterns, financial habits, and operational secrets, creating high-value targets for breaches or profiling. The convenience of automation comes at the cost of potential pervasive surveillance. In such systems, privacy risks are compounded by the erosion of user agency, as continuous monitoring and automated inference leave little room for informed consent, selective participation, or meaningful opt-out mechanisms. Without robust encryption, differential privacy, and strict data governance, this evolution risks turning climate action into digital control. True progress demands not just smarter systems, but more ethical ones [100].

Another implication is the tension between supply chain transparency and confidentiality: blockchain-based tools (e.g., Digital Product Passport, UG-CET, Hyperledger-based trackers) address supply chain traceability and accountability, but their transparency can conflict with corporate confidentiality. While they enhance trust, they may expose trade secrets unless combined with permissioned ledgers or privacy-preserving protocols (e.g., zk-SNARKs). This tension reflects a broader governance issue of how to design transparent yet confidential sustainability reporting systems.

Finally, public calculators and gamified apps, such as WWF, CoolClimate, Tanaffas, and EcoTrack, prioritize engagement and behavior change through simplicity and interactivity; however, they often oversimplify emissions by relying on generalized assumptions, thereby missing critical nuances. In contrast, expert tools like One Click LCA, Greenhouse Gas Protocol, and EC3 deliver high-fidelity, standards-compliant results essential for corporate reporting and policy, but demand technical knowledge and granular data, limiting accessibility. This divide hinders scalable climate action: public tools lack depth; expert tools lack reach. The solution lies in hybrid models: intuitive, user-friendly interfaces powered by sophisticated, behind-the-scenes LCA engines that automate complexity without exposing it. Such designs could empower citizens with accuracy, not just awareness, bridging the gap between engagement and integrity.

B. THREAT MODELING AND ADVERSARIAL PRIVACY RISKS

While the evaluation framework focuses on data collection and processing practices, privacy risk ultimately materializes

TABLE 13. Threat modeling and adversarial privacy risks.

Threat Actor	Capability	Motivation	Affected tool categories
Nation-state or regulator [97]	Traffic analysis, legal compulsion, metadata access	Surveillance, policy enforcement	Cloud-hosted enterprise platforms, centralized dashboards
Corporation or platform provider [32]	Large-scale data aggregation, cross-platform profiling	Behavioral targeting, monetization, competitive advantage	Finance-linked consumer apps, IoT-based tools
Malicious insider [103]	Privileged database or system access	Fraud, coercion, extortion	Centralized organizational tools, consultancy-hosted platforms
External hacker [101, 102]	Exploiting software vulnerabilities, credential theft	Ransom, resale of data	Web-based calculators, poorly secured cloud services

through adversarial actions. Established privacy threat-modeling literature emphasizes that identical data practices can lead to vastly different harms depending on the threat actor, their capabilities, and motivations [101]. Granular or centralized carbon footprint data may therefore expose users and organizations not only to abstract compliance risks, but also to concrete threats such as surveillance, profiling, insider misuse, or external breaches [32]. Drawing on established privacy threat modeling literature and prior work on surveillance, insider threats, and data re-identification, Table 13 summarizes representative threat actors, their capabilities, motivations, and the categories of carbon footprint tools most affected [32], [97], [101], [102], [103].

Prior work demonstrates that fine-grained behavioral or transactional datasets are highly susceptible to re-identification through pattern analysis, even when anonymized [102]. Corporate aggregation of such data further enables behavioral profiling and economic inference, particularly in consumer-facing and finance-linked applications [32], [97]. Centralized sustainability dashboards and cloud-hosted platforms are additionally exposed to insider threats, where authorized access can be abused for fraud, competitive intelligence, or extortion [103]. Finally, web-based calculators and dashboards remain vulnerable to common application-layer attacks documented in established security taxonomies [101], [102].

By making adversarial assumptions explicit, this analysis reinforces that privacy risk is not merely a technical attribute of data collection practices, but a function of who may access the data, under what conditions, and for what purpose. This perspective strengthens the interpretability of the framework and supports more informed tool selection and governance decisions across sectors.

C. THEORETICAL IMPLICATIONS

This study contributes to theory by bridging the fields of sustainability measurement and data privacy management, two domains that have often been studied separately. Existing sustainability research emphasizes the importance of accurate carbon accounting for achieving environmental targets and supporting ESG reporting, while privacy scholarship highlights the ethical and regulatory challenges of data collection. By systematically evaluating carbon footprint tools across five categories, this study integrates these perspectives. It proposes a framework for assessing privacy risk using five criteria: type of data, granularity, storage and processing, potential secondary use, and user agency. By elevating user agency to a core evaluative dimension, the framework extends beyond traditional data-centric privacy models and aligns with GDPR's privacy-by-design principle, emphasizing control and autonomy rather than minimization alone.

The findings reveal that the theoretical relationship between accuracy and privacy is neither linear nor uniform across categories. For example, tools that rely on manual input or aggregated technical data (e.g., WWF Calculator, EC3, Carbon Tracker) demonstrate that meaningful carbon accounting can occur with low privacy intrusion. In contrast, tools that automate data collection through IoT sensors, financial integration, or blockchain immutability achieve higher accuracy but at the cost of greater privacy risks. This suggests that future theories of digital sustainability tools should explicitly account for the trade-offs between informational precision and individual or organizational confidentiality.

Furthermore, the cross-category comparison highlights that privacy risks are context-dependent. Household-focused tools disproportionately affect personal autonomy, while corporate dashboards and blockchain-based systems raise concerns about organizational secrecy and competitive intelligence. This contextual variability implies that privacy in sustainability measurement cannot be treated as a one-size-fits-all concept; rather, it requires a multi-level theoretical lens that differentiates individual, organizational, and systemic risks. Overall, this study advances literature by positioning privacy as a central theoretical construct in the design and evaluation of carbon footprint tools. It opens new pathways for future research to explore privacy-preserving sustainability frameworks, such as federated learning, zero-knowledge proofs, and anonymization, as enablers of responsible digital transformation in sustainability reporting.

D. RESEARCH LIMITATIONS

This study has several limitations that should be acknowledged. First, the analysis is primarily based on a literature-driven evaluation of carbon footprint tools, supplemented by information from professional and industry websites. Accordingly, the classifications should be interpreted as comparative indicators of privacy risk based on available documentation, rather than definitive assessments of deployed system behavior. While this approach ensured

broad coverage across academic and industry sources, it may not fully capture all existing or emerging tools, particularly proprietary systems whose detailed privacy practices are not publicly disclosed. Second, the study employed an exploratory rather than systematic review methodology, which allowed for flexibility in focusing on thematically relevant sources, but may have introduced selection bias by excluding potentially relevant works outside the chosen scope.

Third, the evaluation framework developed in this study, focusing on the type of data collected, granularity, storage, user agency, and potential secondary use, provides a structured lens for privacy assessment. However, privacy is a multidimensional and context-dependent concept, other aspects such as detailed legal compliance assessments, implementation-level consent interfaces, and cybersecurity safeguards were beyond the scope of this paper. Fourth, while the inclusion of industry-grade tools from providers such as SAP, Microsoft, IBM, and Oracle strengthens the practical relevance of the study, the evaluation relied on publicly available documentation and may not reflect the full technical or operational safeguards implemented in practice. Ultimately, the findings highlight general trends across categories rather than providing detailed case studies of tool deployment in specific contexts. As such, while the study identifies comparative privacy risks, the generalizability of results may be limited, highlighting the need for empirical validation through field studies and user-centered investigations.

A further limitation concerns the evaluation of enterprise-grade and proprietary carbon management platforms (e.g., Microsoft, SAP, Oracle, Salesforce). For these tools, privacy risk classifications necessarily relied on publicly available documentation, regulatory disclosures, and secondary technical analyses, rather than direct access to source code or internal system architectures. While this approach is consistent with prior research on commercial digital platforms, it limits the ability to verify claims regarding data minimization, internal data flows, or secondary use practices. To address this uncertainty, the study introduces a confidence rating indicating whether each classification is based on auditable implementations or documentation-based inference. Future research should prioritize technical audits or in-depth case studies with privileged system access to validate and refine these assessments.

Moreover, in the primary analysis, the five evaluation dimensions were treated as equally weighted. This decision reflects a conservative, transparent design choice commonly adopted in early-stage comparative frameworks, where differential weighting may introduce subjectivity or compromise interpretability [101]. However, privacy theory and regulatory practice suggest that not all dimensions contribute equally to real-world risk. In particular, data granularity and secondary use potential often amplify harm more strongly than storage modality alone [97]. For example, continuous high-frequency monitoring (e.g., smart meters) creates substantially greater re-identification and behavioral inference

risk than one-time data collection, even when both involve sensitive domains [102]. Similarly, GDPR places strong emphasis on purpose limitation and downstream data reuse, indicating that secondary use can be more consequential than initial collection context [20].

From a contextual integrity perspective, privacy violations occur not merely through data collection, but when information flows exceed contextual norms, particularly through aggregation, repurposing, or cross-domain linkage [98]. Under this lens, secondary use and granularity represent higher-order risk multipliers, as they enable context collapse even when individual data points appear innocuous [32]. To account for this, a weighted sensitivity interpretation was explored conceptually, in which granularity and secondary use were assigned greater influence than storage method. A representative weighting scheme, consistent with GDPR principles and contextual integrity theory, can be expressed as: Privacy Risk Score = 0.3 (Data Type) + 0.4 (Granularity) + 0.2 (Storage & Processing) + 0.1 (Secondary Use Controls).

Applying such weighting does not alter the relative classification of tools in this study; rather, it reinforces key findings [98]. Tools relying on continuous sensing, financial integration, or immutable ledgers (e.g., smart meters, finance-linked apps, blockchain systems) remain high risk, while tools based on manual input, aggregation, or local processing remain low risk. The primary contribution of weighting is therefore interpretive clarity, not categorical change. Future empirical research could refine these weights through stakeholder elicitation, regulatory benchmarking, or incident-based harm analysis. For the purposes of this study, equal weighting provides a neutral baseline, while the weighted interpretation strengthens theoretical alignment with privacy-by-design and GDPR risk-based regulation.

E. RECOMMENDATIONS AND FUTURE RESEARCH DIRECTIONS

The evaluation of carbon footprint tools highlights an urgent need to balance data accuracy and privacy protection. Based on the analysis, several recommendations can guide policy-makers, tool developers, and organizations. Developers of carbon footprint tools should adopt privacy-by-design principles, embedding safeguards such as anonymization, local processing, and explicit user consent into their systems. Beyond data minimization, privacy-by-design in this context requires preserving user agency through granular consent options, data portability, and enforceable deletion rights, particularly in automated and real-time systems. High-risk tools, particularly those that rely on financial transactions, IoT sensors, or blockchain transparency, should implement privacy-enhancing technologies such as differential privacy or zero-knowledge proofs to limit data exposure. Policymakers and regulators should also establish clear data governance frameworks for sustainability reporting, ensuring that sensitive personal and corporate data are not exploited for secondary purposes. Organizations that adopt third-party

carbon management tools should conduct privacy impact assessments to evaluate risks before integration, particularly when operational or financial data are involved.

The future of carbon footprint lies not in choosing between accuracy and privacy, but in engineering systems that achieve both through advanced cryptographic and distributed technologies. First, federated learning frameworks, such as DP-FedSW, revolutionize collaborative emissions modeling by keeping raw data localized across organizations, factories, utilities, or campuses, while only sharing encrypted model updates to enhance global predictions. This eliminates centralized data hoarding, protects proprietary operations, and enables cross-institutional insights without compromising confidentiality, making it ideal for regulated sectors where data sovereignty is paramount. Second, complementing federated learning frameworks, zk-SNARKs allow entities to cryptographically prove compliance, such as “Scope 3 emissions are under target”, without ever disclosing the underlying data: supply chain sources, energy mix, or production volumes remain hidden, yet verifiable by auditors or regulators. This transforms accountability from a demand for transparency into a guarantee of integrity, enabling secure carbon credit markets and corporate ESG reporting without exposing competitive secrets. Third, differential privacy introduces mathematical rigor to aggregated datasets, adding calibrated noise to prevent re-identification, even when analyzing high-resolution smart meter or transportation patterns. It ensures that urban-scale climate analytics can reveal trends (e.g., district-level heating efficiency) without revealing individual behaviors, protecting vulnerable populations from profiling. Together, these three pillars, federated learning for decentralized collaboration, zk-SNARKs for verifiable secrecy, and differential privacy for statistically safe aggregation, form the foundation of privacy-preserving sustainability. When integrated with blockchain for auditability and IoT for real-time sensing, they enable a new generation of tools that are not only accurate and scalable but also ethically robust. The next frontier isn’t more data, it’s smarter, safer data. Climate action must respect human dignity as much as it reduces emissions.

1) EMERGING TECHNOLOGIES

Beyond incremental improvements to existing tools, emerging privacy-preserving technologies offer transformative potential for carbon footprint measurement systems. Recent advances in cryptography, distributed computation, and decentralized data processing, such as homomorphic encryption, secure multi-party computation, synthetic data generation, and edge computing, provide new mechanisms to reconcile the growing demand for granular carbon data with stringent privacy and data protection requirements. Exploring these technologies represents a critical next step in designing trustworthy, regulation-aware, and scalable carbon accounting infrastructures.

Homomorphic encryption represents a promising direction for privacy-preserving carbon accounting, as it enables

computations to be performed directly on encrypted data without requiring decryption [104]. In the context of carbon footprint tools, this approach could allow organizations to conduct benchmarking, aggregation, or regulatory reporting on emissions data while keeping sensitive operational or financial information confidential [105]. For example, companies could compare Scope 3 emissions performance across supply chains without exposing exact emissions figures or proprietary processes [105]. Although homomorphic encryption is computationally intensive and not yet widely deployed in sustainability tools, recent advances suggest increasing feasibility for real-world applications, making it a valuable avenue for future research in secure carbon analytics [104].

Secure Multi-Party Computation (MPC) enables multiple parties to jointly compute a function over their inputs while ensuring that no participant learns the others' private data [104]. This capability is particularly relevant for multi-organizational carbon accounting, such as sector-wide benchmarking, collaborative Scope 3 reporting, or consortium-based emissions disclosure [106]. By eliminating the need for centralized data pooling, MPC aligns strongly with GDPR principles of data minimization and purpose limitation [106]. Future research could explore the scalability, performance, and governance implications of MPC-based carbon footprint tools, especially in complex supply-chain ecosystems where trust between parties is limited [104], [106].

Synthetic data generation offers another promising approach to mitigating privacy risks by creating artificial datasets that statistically resemble real carbon emissions data without containing identifiable or sensitive records [107]. Such data could be used to train machine learning models, test carbon accounting algorithms, or share benchmark datasets across organizations and research communities [108]. In carbon footprint tools, synthetic data could reduce reliance on real household, financial, or operational data while preserving analytical utility. However, future research is needed to evaluate the fidelity of synthetic carbon datasets, assess re-identification risks, and establish validation standards to ensure that sustainability insights derived from synthetic data remain reliable and policy-relevant [107], [108].

Finally, edge computing shifts data processing from centralized cloud infrastructures to local devices, such as smart meters, building management systems, or on-premise servers [56], [109]. Applied to carbon footprint tools, edge computing can significantly reduce privacy risks by keeping raw data local and transmitting only aggregated or anonymized results [106]. This approach directly supports GDPR principles related to storage limitation and integrity/confidentiality, while also reducing latency and dependency on cloud platforms [106]. Future research should investigate trade-offs between analytical accuracy, system complexity, and privacy protection when deploying edge-based carbon accounting architectures, particularly in smart buildings and urban sustainability systems [109].

2) FUTURE-PROOF PRIVACY

Although this study evaluates privacy risks based on current data practices, privacy should not be understood as a static property of carbon footprint tools [110]. Rather, privacy risks evolve as datasets accumulate, analytical capabilities advance, and new data sources become linkable. For example, historical smart meter or building energy data may initially appear anonymized and low risk, yet become re-identifiable when combined with future mobility records, social media data, or leaked commercial databases [111]. As carbon footprint tools increasingly rely on longitudinal data collection, the cumulative and retrospective privacy impact may exceed initial assessments.

This concern is particularly acute for blockchain-based carbon tools, where immutability conflicts with fundamental privacy principles such as the right to erasure and storage limitation [111]. Data recorded today may persist indefinitely, even as regulatory requirements, social norms, or user expectations change. In such contexts, transparency and auditability must be balanced against long-term exposure risks.

To address these challenges, future carbon accounting systems should adopt future-proof privacy strategies, including sunset clauses for data retention, periodic risk re-evaluations, and adaptive governance mechanisms [110]. Limiting data retention duration, enforcing automatic deletion or aggregation over time, and separating raw data from long-term records can reduce cumulative exposure [111]. Embedding temporal privacy considerations into system design aligns with GDPR's risk-based and lifecycle-oriented approach and is essential for ensuring that sustainability tools remain trustworthy as technological and societal contexts evolve [110].

3) NUANCED LIMITATIONS OF PRIVACY-PRESERVING TECHNOLOGIES IN CARBON ACCOUNTING

While differential privacy is frequently presented as a robust solution for protecting sensitive emissions data, its effectiveness depends critically on the choice of the privacy budget (ϵ). In practice, there is no universally "acceptable" ϵ value for carbon reporting, as privacy, utility trade-offs vary by sector and data sensitivity [112]. Prior work in energy analytics and mobility datasets suggests that ϵ values in the range of 0.5–1.0 can preserve aggregate trends while limiting re-identification risk, whereas stronger guarantees (e.g., $\epsilon \leq 0.1$) often result in substantial utility degradation, reducing the reliability of fine-grained emissions estimates [112]. This trade-off implies that DP is more suitable for macro-level reporting (e.g., city, sector, or portfolio averages) than for high-resolution household or firm-level accounting [82]. Sector-specific guidance, such as energy data anonymization standards and mobility privacy frameworks, should therefore inform ϵ selection in sustainability contexts rather than adopting arbitrary thresholds [113].

Moreover, zero-knowledge proofs, particularly zk-SNARKs, offer strong confidentiality guarantees by

enabling entities to prove compliance with emissions targets without revealing underlying data [55]. zk-SNARKs are known for their short verification times, typically adding around 10-100ms latency [114]. This latency is generally acceptable for many applications, including real-time carbon markets, due to the efficiency and scalability of zk-SNARKs. However, their deployment entails non-trivial assumptions and trade-offs. Many zk-SNARK constructions require a trusted setup ceremony, the compromise of which could undermine system integrity [114]. Additionally, while verification costs are relatively low, typically on the order of tens to hundreds of milliseconds, this overhead may constrain applicability in latency-sensitive environments such as real-time carbon trading or high-frequency reporting platforms [115]. Importantly, zero-knowledge proofs verify the consistency of reported claims with declared rules, not the factual correctness of the underlying measurements; inaccurate or manipulated input data can still produce valid proofs [116]. As such, zero-knowledge proofs are best viewed as complementary governance tools that enhance confidentiality and auditability, rather than as standalone guarantees of data quality [117].

Federated learning architectures, including DP-FedSW, reduce privacy risks by keeping raw emissions data localized while sharing model updates [118]. Nevertheless, federated learning is not immune to privacy leakage. Prior studies have demonstrated the feasibility of model inversion and membership inference attacks, particularly in settings with few participants or insufficient noise injection [119]. Empirical evidence suggests that federated deployments with a small number of participants (e.g., fewer than several dozen entities) may still enable reconstruction of sensitive attributes from gradients. Furthermore, gradient sharing without proper clipping, aggregation, or differential privacy can leak information about individual contributors [120]. These limitations indicate that federated learning-based carbon accounting systems must incorporate additional safeguards, such as participant thresholds, secure aggregation, and calibrated noise, to achieve meaningful privacy protection.

Collectively, these considerations highlight that privacy-preserving technologies are not plug-and-play solutions; their effectiveness in carbon accounting depends on careful parameterization, governance design, and contextual deployment rather than purely technical adoption [121], [122].

VII. CONCLUSION AND FUTURE RESEARCH DIRECTION

As carbon accounting becomes central to addressing climate change and supporting ESG reporting, a growing number of digital tools are being developed to measure individual, organizational, and systemic carbon footprints. However, the very data that makes these tools effective, ranging from household behaviors to financial transactions and real-time sensor readings, can also threaten user privacy. This tension between accuracy and confidentiality remains underexplored, creating a critical research gap. This paper contributes to

the literature by systematically evaluating carbon footprint tools across five categories, utilizing a structured privacy framework that considers data type, granularity, storage and processing, potential secondary use, and user agency. The cross-category analysis identified both the least risky tools, which demonstrate privacy-preserving practices, and the riskiest tools, which highlight urgent vulnerabilities. The findings reveal pronounced privacy-accuracy trade-offs: tools offering higher measurement precision often require more intrusive data collection. The study also singles out the lowest- and highest-risk categories of tools, noting that basic household calculators pose minimal risk. In contrast, enterprise platforms and IoT/blockchain systems face the greatest exposure because they handle sensitive operational, financial, and real-time data. The findings show that privacy risks are context-dependent, rising significantly in tools that rely on automation, financial integration, or immutable ledgers. By explicitly bringing privacy and user agency into the evaluation of carbon footprint tools, this study offers both theoretical and practical insights. It emphasizes that sustainability measurement must be pursued in conjunction with robust privacy safeguards.

FUTURE RESEARCH AGENDA

Building on the insights of this study, several concrete research directions emerge. These research questions point toward a broader agenda aimed at advancing privacy-preserving carbon accounting, bridging the gap between environmental data needs and the ethical management of sensitive information.

RQ1: What differential privacy epsilon values provide acceptable accuracy-privacy trade-offs for corporate Scope 3 reporting?

RQ2: How do users' mental models of carbon tool privacy align with actual data practices?

RQ3: Can synthetic carbon data enable model training without real-world data exposure?

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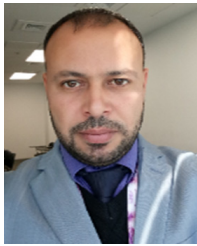
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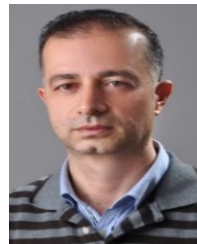
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